SPEAR3 LLRF Upgrade System

System Design Report

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This report is the top-level design reference for the SPEAR3 LLRF Upgrade Project. It describes the system architecture, the scope and high-level design of every subsystem in both the legacy and upgraded configurations, and the interfaces between them. It is the entry point for the detailed engineering design of each subsystem, not a substitute for it: §20 indexes the detailed design documents that carry the underlying evidence.

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4 Arc detection signal path. The dashed outline is the vendor boundary: everything inside it is AFT ARC4 hardware, and no SSRL wire lands on an individual detector module. Fibers enter the detector modules from the front; the detector modules drive the chassis backplane; only the two system modules and the test module face the Interface Chassis. Ten of the chassis's twelve slots are populated; slots 11 and 12 are left empty and their two detector modules are held as shelf spares. 36
5 Klystron cathode heater controller architecture. Three paths are deliberately separate: the MPS ramps the TDK Lambda output through the *analog* programming input, so controlled warm-up and cooldown do not depend on the network; the Ethernet link carries setpoints and readback only; and the existing analog V/I meters are retained, so the delivered output is monitored in hardware independently of both. The external $60\,\mathrm{Hz}$ sync circuit is a new in-house design item. There are no mechanical components anywhere in the control path. . 39
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Revision History

Revision	Date	Scope of Change	Summary
R4	2026-11-05	RF Local Display Panel; beam abort; tuner control; HVPS controller; transition; status	**RF Local Display Panel**: the Local Control Chassis and the Local Panel 340-310 are one unit, named after its drawings; its three HVPS fibers run to the HVPS controller in B118 and its own logic forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort to the SPEAR MPS (J. Sebek's fiber note and the panel drawings). **Beam abort**: kept as an Interface Chassis output. **Tuners**: LLRF9 Unit 1 runs the four tuner loops and drives the Galil DMC-4143 directly. **LLRF9 cabling**: the channel tables give the current cabling of the master (Unit 1, the vendor's four-cavity configuration, as in the 2021 setup) and the slave (Unit 2: reflected and load signals); the klystron drive power, on neither unit, is proposed for Waveform Buffer channel 1. **Status**: heater supply on a test stand with the live filament limits recorded; PPS Interface Box awaiting ADSO approval. **EPICS IOC**: the coordinator is the IOC B132-IOCRF, which moves from the VXI CPU to new hardware when the crate is retired. **HVPS controller**: a complete new Hoffman box for each HVPS unit retires the existing one; all HVPS control is in EPICS, with no Python layer, under the current HVPS PV names. **Transition**: a new section gives the integration sequence I0–I8 of the Subsystem Integration and Migration Plan (Doc M); the Interface Chassis section adds the transition interfaces it needs while the AIM, the PLC-5 and the legacy HVPS controllers are still in service, and the command path that replaces the AIM cable J15; a table gives the new home of each legacy function that has no one-to-one successor. **Open items**: O1 (soft IOC location) and O8 (tuner control split) closed and the others renumbered O1–O7; O3 now covers the panel logic and the beam-abort endpoint, and O4 the Waveform Buffer channel assignment. **Corrections**: the 10:1 T1 of the heater chassis is the ammeter's current transformer, not a step-down transformer; the tuner motors have no encoders; the legacy automatic reset is inhibited by the AIM forced-fault latch.

Revision	Date	Scope of Change	Summary
R3	2026-09-25	Arc detection re-selection; legacy unit disposition; chassis consolidation; heater vendor	**Arc detection**: Microstep-MIS abandoned (vendor unresponsive, and it would have required custom CF-flange adapters and a viewport redesign); replaced by the **AFT microwave GmbH ARC4** vendor chassis, whose fiber inputs mate with the tunnel fibers already installed — no mechanical work. The custom SSRL Arc Detection Chassis OR/latch design is withdrawn; the Interface Chassis now faces only the 2 ARC4 system modules and the test module. Figure 4 redrawn. **Legacy units**: Fast Interlock Chassis 340-308, AIM (VXI Slot 12) and Local Panel 340-310 named explicitly as removed with the VXI crate, their functions absorbed by the Interface Chassis. **Chassis consolidation**: the Waveform Buffer and Interface Chassis share one physical chassis on two PCBs; Waveform Buffer complete (April 2026), Interface Chassis PCB at fabrication since August 2026. **Heater**: Chroma → **TDK Lambda**, with an external 60 Hz sync circuit in design.
R2	2026-04-28	Technical consolidation and drawing update	Finalized system architecture and protection chain. Expanded and corrected arc detection design, refined interface-chassis / MPS responsibilities, corrected cavity gap voltage values, confirmed heater controller approach, and standardized terminology and signal naming across subsystems.
R1	2026-03-17	Design clarification and corrections	Incorporated review feedback from J. Sebek: clarified HVPS power chain and signal flow, corrected RF signal monitoring assignments, refined software architecture descriptions, and resolved ambiguities in interlock and protection narratives.
R0	2026-03-03	Initial release	Baseline physical design report documenting the legacy SPEAR3 RF system and the proposed full LLRF upgrade architecture. Established subsystem breakdown, retained vs. replaced hardware, and initial protection philosophy.

Open Items — Still To Be Confirmed

Everything in this report that is **provisional, unverified or explicitly TBD** is collected here. Each item is tagged **[O*n*]** at the point in the body where it matters; the page column links back to that point.

ID	Item	What is still needed	Needed before	Page
O1	LLRF9 ↔ Interface Chassis logic levels	Exact voltage levels for the interlock input and output. The input is selectable (3.3 / 5 / 24 V); the output is specified as 4.75 V high through 220 Ω. Both Interface Chassis stages must be designed to match.	Interface Chassis PCB design	p. 19
O2	PPS connector part number	Field verification: the original Burndy circular 8-pin may already have been replaced by the Souriau Trim Trio equivalent (GOB1208PNE).	PPS Interface Box fabrication	p. 22

ID	Item	What is still needed	Needed before	Page
O3	RF Local Display Panel logic and beam-abort endpoint	The panel's I/O is documented in its drawings and in J. Sebek's fiber note, but the equations of its programmable logic device (GAL22V10), which forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort, are not. The endpoint of its beam-abort output (rear BNC J6) in the SPEAR MPS and the level that input expects are not on any RF drawing.	Interface Chassis logic design	p. 28
O4	Waveform Buffer RF channel assignment	Channel 1 is proposed for the klystron drive power, which is on neither LLRF9 unit. Channels 7 and 8 are unassigned; one is a candidate for a coupled klystron forward power signal feeding the collector power calculation.	Waveform Buffer channel assignment	p. 33
O5	AFT ARC4 $\rightarrow$ Interface Chassis	The external interface of the two system modules and the test module is not specified from vendor documentation: number of discrete lines, logic levels, whether fired-detector identification is parallel bits or an encoded word, and the latch reset mechanism.	Interface Chassis PCB layout	p. 36
O6	Klystron heater supply	The TDK Lambda unit is installed on a test stand. Its ASCII command subset, regulation figure and MTBF figure are not yet recorded here; the ASYN EPICS module does not exist yet; the external 60 Hz sync circuit is in design. All are to be closed in an acceptance test on the test stand with a load.	Heater subsystem commissioning	p. 38
O7	Network addressing	IP addresses and subnet allocation for both LLRF9 units, both PLCs, the motion controller and the Waveform Buffer are all unassigned.	Any integration testing	p. 48

Purpose and Scope

This Physical Design Report is the top-level system design reference for the SPEAR3 Low-Level RF (LLRF) Upgrade Project. It describes the overall system architecture, the scope and high-level design of each subsystem in both the current (legacy) and upgraded configurations, and the interfaces and interactions between subsystems.

This document is intended to serve as the entry point for the detailed engineering design of each subsystem. It consolidates physical design information from across the project and provides traceability to the detailed design documents in this project.

1 Executive Summary

The SPEAR3 RF station provides 476.3 MHz RF power to the SPEAR3 storage ring at the Stanford Synchrotron Radiation Lightsource (SSRL). A single klystron, driven at approximately 800 kW, feeds four single-cell RF cavities through a waveguide distribution network. The combined cavity gap voltage is approximately 2.85 MV.

The LLRF Upgrade Project replaces the entire control electronics chain — not merely the low-level RF controller, but also the RF machine protection system (RF MPS), HVPS controller, tuner motor controllers, and supporting infrastructure. The project will also modify several systems in the existing system. An Interface Chassis that is required to centralize and coordinate the interlocks between system elements, will be redesigned and replaced to interact with the elements of the system. The klystron cathode heater power supply will be modernized. An expanded Waveform Buffer System is being built to provide better system diagnostic signals. And a new optical arc detection system will be installed to replace a system that did not perform as expected.

1.1 Key System Parameters

Parameter	Value
RF Frequency	~476.3 MHz
Total Gap Voltage	~2.85 MV (sum of 4 cavities)
Cavity Gap Voltage	~712 kV each
Klystron Power	~1.5 MW rated, ~800 kW typical operating
HVPS Voltage	up to ~90 kV (negative polarity)
HVPS Operating Voltage	~74 kV at 500 mA beam
Drive Power	~29 W nominal
Number of Cavities	4 (single-cell, individual tuners)
LLRF9 Units	2 active, 2 spare (4 purchased)

Source: Gap voltage and beam current data from LLRF9 commissioning measurements (`llrf/tests/llrf9Tests.pdf`, J. Sebek, 2021). Klystron rated power from HVPS hazard documentation (`hvps/documentation/procedures/spear3HvpsHazards.tex`).

1.2 Upgrade Drivers

- **Hardware obsolescence** — Commercial VXI, PLC-5, and SLC-500 modules are at end of life and no longer vendor supported. Custom SLAC LLRF processing modules are also end-of-life and no longer supported by SLAC.
- **PPS compliance** — Legacy design routes PPS wiring through HVPS controller and has a PLC in the safety chain
- **Performance improvement** — Digital FPGA-based feedback (270 ns loop delay) replaces analog processing
- **Diagnostics** — 16k-sample waveform capture, circular buffers, first-fault detection for RF signals, 100 ms of waveform capture for HVPS signals
- **Maintainability** — Modern hardware, comprehensive documentation, EPICS/MATLAB software

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2 System-Level Architecture

2.1 Legacy System Architecture

The legacy SPEAR3 LLRF system was originally designed for the PEP-II B-Factory (circa 1997) and later adapted for SPEAR3. It consists of:

- **LLRF Controller**: Custom PEP-II analog RF Processor (RFP) module in a VXI chassis. The VXI crate contains a CPU, an Allen-Bradley controller, a Clock Module, the RFP module (heart of the fast feedback system), three IQA modules (amplitude/phase monitors), and the Arc/Interlock Module (AIM, Slot 12).
- **Control Software**: Six SNL (State Notation Language) programs on VxWorks RTOS, compiled into a single `rfSeq` IOC library:
  - `rf_tuner_loop.st,v` — Cavity tuner stepper motor control
  - `rf_hvps_loop.st,v` — HVPS supervisory control and regulation
  - `rf_states.st,v` — RF station state machine control (PEP-II heritage, R. Sass, 1997)
  - `rf_dac_loop.st,v` — Drive Power and Gap Voltage RFP/GFF DAC loop (S. Allison, 1997)
  - `rf_calib.st,v` — Calibration sequences (R. Claus, SLAC/PEP-II LLRF Group)
  - `rf_msgs.st,v` — Message logging and CAMAC TAXI error monitoring
  - See: `spear-rf-code-legacy/rfApp/src/seq/Makefile,v`, `spear-rf-code-legacy/rfApp/src/seq/rf_dac_loop_pvs.h,v`
- **HVPS Controller**: Allen-Bradley SLC-500 PLC (processor: AB-1747-L532, scanner: AB-1747-DCM) with Enerpro SCR gate driver boards, housed in a Hoffman NEMA enclosure in Building B118. It communicates with the VXI crate via an Allen-Bradley serial interface to receive commands and to send readbacks and statuses. It communicates with an interface chassis via fiber optic signals to receive hardware permits and send a hardware status.
  - See: `pps/HoffmanBoxPPSWiring.docx`
- **RF Machine Protection System (RF MPS)**: Allen-Bradley PLC-5 (1771 series), (since converted to ControlLogix 1756 with the hardware assembled, software written, tested without RF power). Scope: klystron/RF station equipment protection only — distinct from the facility-wide SPEAR MPS which provides an external permit input.
  - See: `llrf/documentation/mpsWiringDiagrams/`, `hvps/architecture/designNotes/RFSystemMPSRequirements.docx`
- **Tuner Motor Controllers**: Allen-Bradley 1746-HSTP1 stepper modules with Superior Electric SS2000MD4-M Slo-Syn PWM drivers (obsolete); stepper motors are Superior Electric Slo-Syn M093-FC11 (NEMA 34D)
  - See: `llrf/tuners/SLO-SYN_SS2000MD4M_Step_Drive_Translator_Manual.pdf`, `llrf/tuners/SLO-SYN.pdf`
- **Interlock System**: Distributed across analog modules, PLC I/O, and direct wiring. The Fast Interlock Chassis 340-308 has fiber optic inputs to detect potential arcs in the waveguide system and reflected-power comparators. It summarizes the status of these interlocks and reports them back to the VXI system through the Arc/Interlock Module (AIM, drawing 340-307), located in Slot 12 of the VXI crate, and passes the AIM's commands over fibers to the **RF Local Display Panel 340-310** (rack B132-12). The panel connects part of the hardwired interlock and control wiring to the RF MPS PLC, provides the operator-accessible local control and indication, and terminates the three HVPS fibers: its own logic forms SCR ENABLE and KLYSTRON CROWBAR for the HVPS controller in B118 and a beam abort for the SPEAR MPS (J. Sebek's fiber note; drawings in `llrf/documentation/localPanel/`).
- **Communication**: The main intelligence for the existing system is a Kinetics Systems VXI IOC in the system VXI crate. The crate has a VME Allen-Bradley communication module through which it communicates, via a serial link, with the two Allen-Bradley RF MPS crates, the four cavity tuner stepper motor controllers, and the HVPS controller.
  - See: `spear-rf-code-legacy/rfApp/src/seq/rf_msgs.st,v`

2.2 Upgraded System Architecture

The upgraded system replaces all control electronics while retaining the RF plant physical infrastructure (klystron, cavities, waveguide, HVPS power section, tuner mechanical assemblies). The upgraded architecture is:

Key Architectural Principle: The Interface Chassis implements **machine/equipment protection and operational interlocks** (LLRF/HVPS/RF MPS coordination), while personnel safety (PPS) functions are implemented exclusively in a completely separate, dedicated **PPS Interface Box**. These two safety-related subsystems are architecturally independent:

- **Operator Layer**: has relative little control with only necessary readbacks.
- **Expert Layer**: with details used by system experts to control and read data from the system. There will be MATLAB tools, written by Dmitry for the LLRF9 and written in-house for other systems.
- **HVPS Power section** has three fiber optic cables between Interface Chassis.
- **Heater Control**: will be controlled by RF MPS system.
- **Interface Chassis (Machine/Equipment Protection)**: First-fault detection, optocoupler and/or other galvanic isolation, fiber I/O for LLRF9/HVPS/RF MPS/orbit coordination and other machine-protection interlocks
- **PPS Interface Box (Personnel Safety / PPS)**: Separate Bud enclosure with 4 relays, status LEDs, PPS-lockable connector for HVPS contactor relay and Ross switch control

2.3 What Stays, What Changes, What Is New

Retained Physical Hardware (same equipment):

- Klystron and its RF output

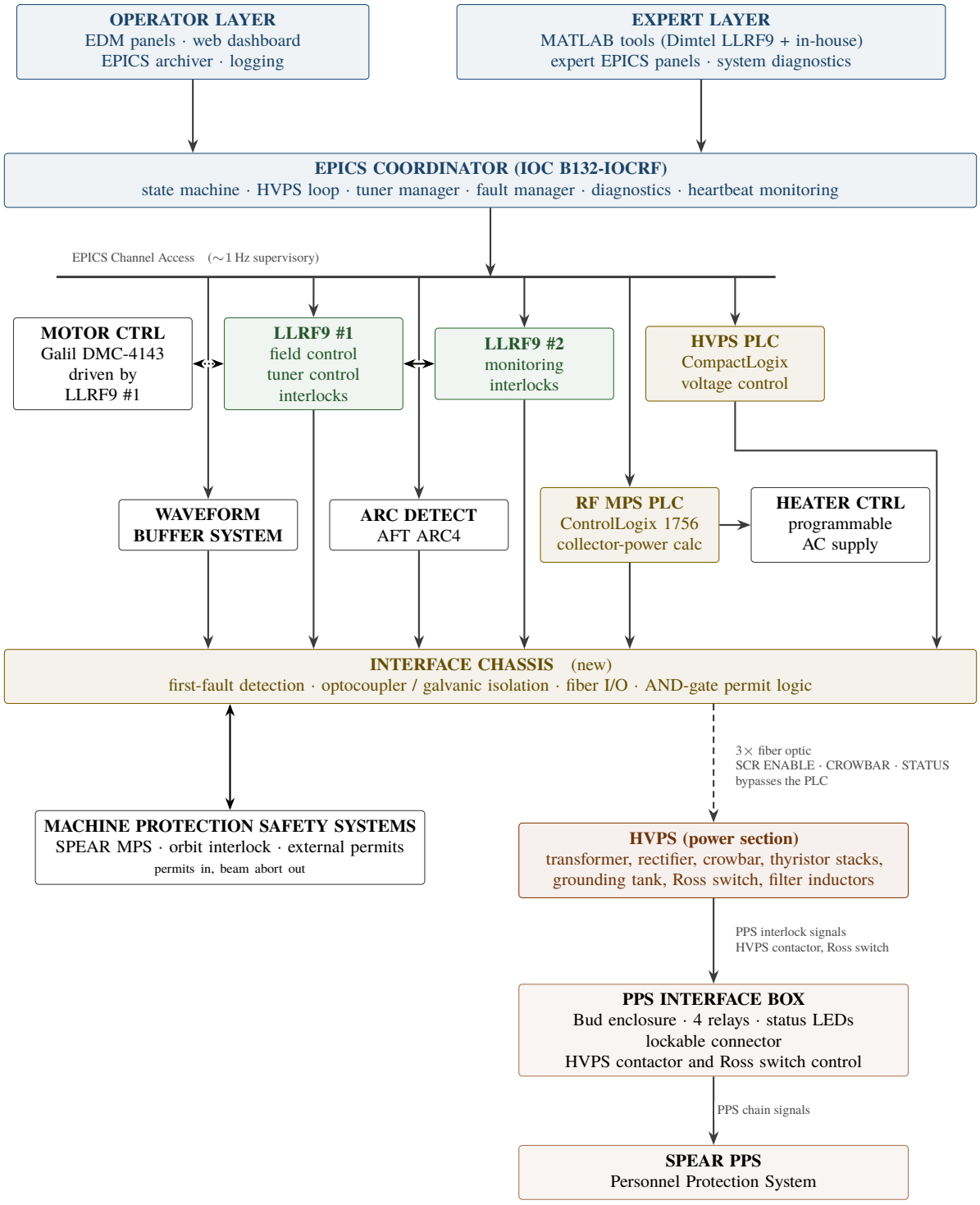


Figure 1: Upgraded system architecture. The Interface Chassis is the single hub for machine and equipment protection; personnel safety runs on a physically and logically separate path through the PPS Interface Box to the SPEAR PPS.

- 4 RF cavities and waveguide distribution (circulator, magic-tees, waveguide loads)
- Stepper motors (M093-FC11 or equivalent) and mechanical tuner assemblies
- Linear potentiometers on tuners (position indication)
- HVPS power electronics (transformer, rectifier, oil system, crowbar)
- Vacuum contactor (Ross HQ3) and contactor controller (Ross HCA-1-A)
- Ross grounding switch, Danfysik DC-CT, Pearson CT-110
- Field cabling (Belden 83715 15C #16 Teflon, Belden 83709 9C #16 Teflon)

Removed / Decommissioned (retired outright — no replacement in kind; their functions move into the Interface Chassis):

- **VXI crate** (Elma 13-slot) and every module in it — CPU, Clock Module, RFP, IQA $\times 3$ and the AIM — and with it the Allen-Bradley Remote I/O serial link that the Slot 1 scanner mastered. The EPICS IOC B132-IOCRF, which runs on the VXI CPU today, moves to new hardware when the crate is retired.
- **Fast Interlock Chassis 340-308** — the legacy fast trip-decision element. Its fiber-optic arc inputs and reflected-power comparators are absorbed by the Interface Chassis, together with the fibers that link it to the RF Local Display Panel.
- **AIM — Arc/Interlock Module** (VXI Slot 12, SLAC drawing 340-307), the digital companion to the Fast IC. Its per-channel arc latching and `RF_FAULT` backplane assertion are replaced by the Interface Chassis first-fault register; its fault-history ring buffer is replaced by the Waveform Buffer circular buffers.
- **RF Local Display Panel 340-310** (rack B132-12; also referred to as the Local Control Chassis) — legacy local/remote control and indication panel and the end point of the three HVPS fibers; its logic forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort to the SPEAR MPS (drawings in `llrf/documentation/localPanel/`). These functions move to the Interface Chassis; local control moves to the Interface Chassis front panel and to EPICS.

Replaced / Upgraded:

- LLRF Controller: Analog RFP → Dimtel LLRF9/476 ($\times 2$ units)
- RF MPS: PLC-5 1771 → ControlLogix 1756
- HVPS Controller: the legacy Hoffman box (SLC-500, Enerpro firing boards, regulator card, trigger interconnect boards, power supplies, local control panel) → a complete new Hoffman box for each HVPS unit, with a CompactLogix PLC, Enerpro FCOG1200 boards and a redesigned analog regulator
- Tuner Motor Controllers: AB 1746-HSTP1 + Slo-Syn (**in service**) → Galil DMC-4143 Rev 1.3h 4-axis motion controller, driven directly by LLRF9 Unit 1 (**built and tested on the SPEAR3 tuner motors under an EPICS control by M. Dunning; not in service; goes online when the LLRF upgrade completes**)
  - See: `llrf/tuners/galil/functioningGalil20250825SwapABToManual.txt`, `llrf/tuners/galil/firstMotion2024.txt`
- Control Software: SNL/VxWorks → EPICS + LLRF9 internal EPICS IOC + MATLAB applications
- Operator Interface: Legacy EDM panels → modernized panels

New Subsystems (did not exist in legacy):

- Interface Chassis — central hardware interlock coordination hub
- Waveform Buffer System — 8 RF + 4 HVPS channel extended monitoring (shares one physical chassis with the Interface Chassis, on a separate PCB)
- Arc Detection — **AFT microwave GmbH ARC4** vendor chassis with 12 optical detector modules: 8 cavity window channels (2 per cavity), 1 klystron output window, 1 circulator, 2 spares. Existing tunnel fibers will be test and replaced as needed; no viewport or mechanical adapter work is required.
- PPS Interface Box

Enhanced Subsystems (upgraded from legacy):

- Klystron Heater Controller — Motor-driven variac → **TDK Lambda** programmable AC supply with an external 60 Hz sync circuit; MPS controls via analog ramp, EPICS monitors/enables over Ethernet.
- Collector Power Protection — Forward power proxy → Waveform Buffer System measures HVPS voltage and current and klystron forward power → Collector power calculation performed in RF MPS

2.4 Legacy Functions Without a One-to-One Successor

Several functions of the legacy station belong to none of the replaced units alone. Each has a home in the upgrade; the order in which they move is part of the integration sequence (Section 19.2).

Function	Legacy implementation	Home in the upgrade
Beam abort to the SPEAR MPS	Forced and reset by the state machine through the AIM; the RF Local Display Panel adds the HVPS STATUS and the solenoid faults; rear BNC J6	Interface Chassis output; force and reset from the coordinator

Function	Legacy implementation	Home in the upgrade
Main focus and bucking-coil supply control and interlocks	RF MPS PLC-5 through the panel	RF MPS outputs
Filament request and timer bypass	AIM, Fast Interlock Chassis, panel and PLC-5	RF MPS and the heater IOC
HV-detected, RF-detected, local/remote and online/offline indications	Panel to the PLC-5	RF MPS inputs
HVPS voltage and current for the collector power	Buffered by the panel for the PLC-5	Analog signals from the HVPS controller to the Waveform Buffer and the RF MPS
HV turn-on gate	AIM "HVPS on" request into the panel's logic	Interface Chassis logic; the request from the coordinator
Automatic reset after a trip, with its retry count and forced-fault inhibit	SNL state machine; AIM forced-fault latch, set by operator command	Coordinator
Reference power, crate and Remote I/O health monitors	SNL and VXI records	Reference distribution monitor; IOC health records
Fault capture and MATLAB fault tools	AIM, IQA and RFP history buffers; MATLAB tools	LLRF9 and Waveform Buffer post-mortem; ported tools
Readbacks for other subsystems (tuner positions and temperatures, stepper supply, filament, coil supplies, ion pumps, window temperatures, HVPS oil temperature)	PLC-5 channels	RF MPS, under the same PV names
476 MHz reference distribution	VXI CLK module	Separate reference distribution chassis

3 Physical Layout and Locations

The SPEAR3 RF station spans multiple buildings and locations at SSRL:

Location	Equipment	Notes
Building B118 (Power Supply Room)	HVPS Controller (Hoffman Box)	HVPS control location
Building B514 (HVPS Substation)	HVPS Main Tank (transformer, rectifier, inductor, filter caps), Phase Tank (12 thyristor stacks), Crowbar Tank (4 thyristor stacks, output V-divider)	High-voltage power section; FR3 oil-filled, N_2 blanket
Contactor Disconnect Panel (Switchgear, B514)	Vacuum contactor (Ross HQ3), Contactor controller (Ross HCA-1-A), K4/MX/RR/L1 relays, S5 auxiliary contact	12.47 kV AC switchgear
Termination Tank (B132, near klystron)	HV cable termination, Ross Engineering HV relay, Danfysik DC-CT, Pearson CT-110	Mineral oil filled
Switch-over Tank (adjacent B514)	HV cable connections between HVPS1/HVPS2 and klystron	FR3 oil filled
Building B132 (Klystron)	Klystron, drive amplifier, LLRF9 units, RF MPS PLC, combined Interface Chassis / Waveform Buffer chassis, Motion Controller, AFT ARC4 arc detection chassis, EPICS IOC B132-IOCRF (moved from the VXI CPU to new hardware)	Main control electronics location
SPEAR3 Storage Ring Tunnel	4 RF cavities, waveguide distribution, tuner assemblies, arc detection sensors	Radiation area

3.1 Cabling Between Locations

Cable Run	Cable Type	Conductors	Route
B118 → Switchgear (Contactor)	Belden 83715	15C #16 Teflon	TS-5 to contactor controller
B118 → Termination Tank (Grounding)	Belden 83709 + Belden 83715	9C + 15C #16 Teflon	TS-6 to grounding tank
B118 → B514 (HVPS)	Electrical cable pairs	SCR trigger cables (12 pairs)	Controller to Phase Tank thyristor stacks
B132 ↔ B118 (HVPS controller)	Fiber optic	SCR ENABLE, KLYSTRON CROWBAR, STATUS	RF Local Display Panel ↔ trigger interconnect boards of the HVPS controller (upgrade: Interface Chassis ↔ HVPS controller)
B132 → Tunnel	Coax cables	RF signals (forward, reflected, probe). Similar forward, reflected power from klystron and circulator load with B132	LLRF inputs from cavities
B132 → Tunnel	Multi-conductor	Motor power and limit-switch signals	To tuner assemblies

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4 RF Plant — Retained Physical Infrastructure

The RF plant is the physical chain that delivers 476.3 MHz RF power from the klystron to the storage ring beam. All of this hardware is retained as-is during the upgrade.

4.1 Klystron

The single klystron is located in Building B132 and operates at approximately 800 kW output power, driven at ~29 W input from the drive amplifier. The klystron cathode is powered by the HVPS at up to ~90 kV (nominal ~74.7 kV at 500 mA beam current). It has a non-full-power collector requiring dedicated protection (see Section 11.3 for upgrade and current implementation in Section 4.5).

4.2 Waveguide Distribution

The klystron output feeds a waveguide network consisting of:
- A circulator (protects klystron from reflected power)
- Magic-tee power splitters (two stage splitter that distributes power to 4 cavities)
- Waveguide loads (absorb rejected power)

4.3 RF Cavities

Four single-cell cavities at 476 MHz, each contributing ~712 kV gap voltage for a total of ~2.85 MV. Each cavity has:
- A cavity probe (monitors internal field amplitude and phase)
- A forward power coupler
- A reflected power coupler
- An individual stepper motor tuner (adjusts resonant frequency via mechanical plunger)
- Waveguide window viewports on either side of a ceramic window (for arc detection sensor mounting)

4.4 Drive Amplifier

The drive amplifier is located in Building B132 near the klystron and boosts the LLRF9 output signal (~0 dBm) to ~29 W to drive the klystron input. In the upgrade, the LLRF9 Unit 1 output (from the thermally stabilized output chain on Board 1 or Board 2) connects to the drive amplifier input via coax cable. A KAW2051M12 amplifier datasheet is available in `llrf/driveAmp/`.

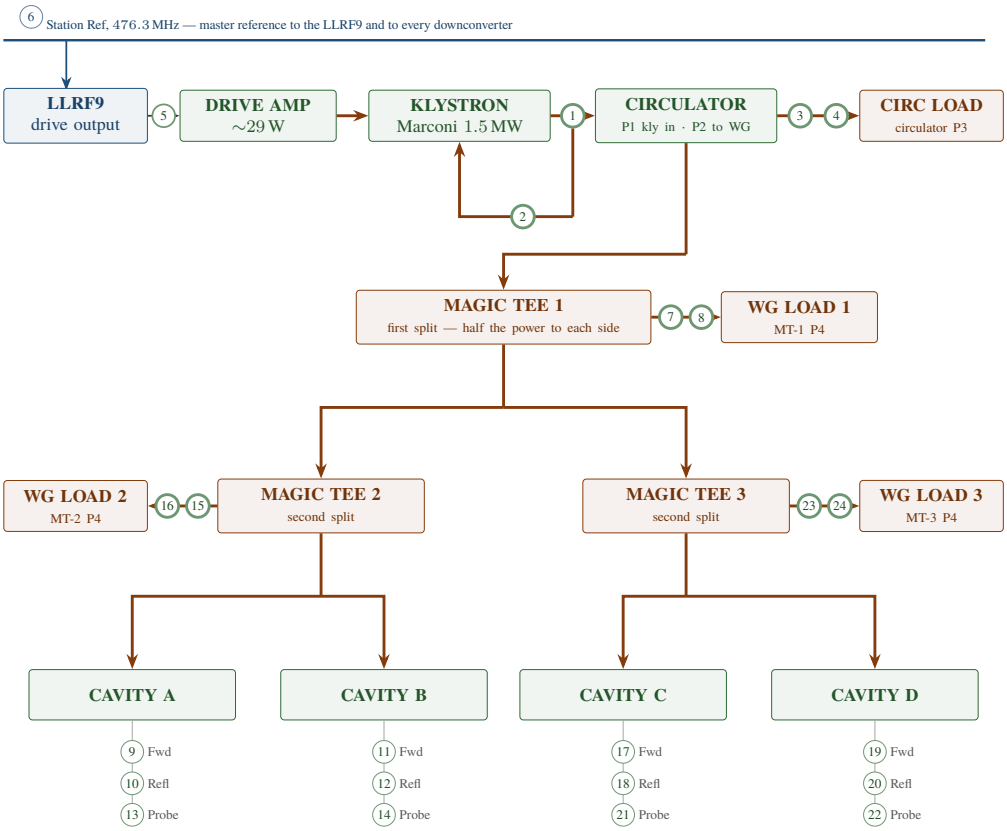


Figure 2: RF plant signal flow. Heavy lines are waveguide, light lines coaxial. Circled numbers are the directional-coupler channel indices used in §4.6. Channel 2 (klystron reflected power) branches from the bottom of channel 1 because both are ports of the same directional coupler between the klystron and the circulator.

4.5 Legacy Collector Power Protection

The current SPEAR3 system includes collector power protection implemented through the HVPS control loop. The legacy system monitors klystron forward power and compares it against configurable limits:

Current Implementation:

- **Monitoring**: `LLRF:HVPS:PCOLL_MAX` PV sets maximum collector power limit
- **Protection Logic**: HVPS loop (`rf_hvps_loop.st,v`) monitors `klystron_forward_power` vs. `max_klystron_forward_power`
- **Action**: When klystron forward power exceeds limits, HVPS voltage is reduced (`delta_proc_voltage_down`)
- **Integration**: Collector limit displayed in HVPS EDL panel ("COLLECTOR LIMIT")

> **Note**: In addition to the software protection above, there is an MPS limit (responding on the order of ms) that removes the permit from the HVPS if the limit is exceeded. The fast interlock chassis has an internal RF detector module that takes a coupled signal from the klystron forward power. The AB controller monitors HVPS V and I and performs the arithmetic, removing the MPS permit if collector power exceeds the allowable value. Removal of the permit turns off both the HVPS and the drive power.

Legacy Limitations:

- No direct DC power measurement

4.6 RF Signal Monitoring

The amplitudes of the RF data need to be acquired. The RF signals that are currently monitored are listed in the table below, along with their location in the RF plant signal flow.

4.6.1 Monitored RF Signals

#	Signal	Monitored By	Notes
1	Klystron Output Forward Power	LLRF9 Unit 1 CH2	RF power at the klystron output traveling toward the circulator; primary measure of station RF output level.
2	Klystron Output Reflected Power	LLRF9 Unit 2 CH4	RF power reflected back into the klystron output port from the load chain; should be near zero; used for klystron protection.
3	Circulator Load Forward Power	LLRF9 Unit 2 CH5	Power delivered to the circulator load (circulator port 3); absorbs power the circulator does not properly direct to the cavities plus any net reflection returning from the waveguide network. Sits at ~20 kW, increases to ~40 kW at full power with no beam — more significant signal than WG Load 1 Fwd.
4	Circulator Load Reflected Power	Waveform Buffer Ch2	Power reflected from the circulator load back into the circulator; indicates load match quality.
5	Klystron Drive Power	Waveform Buffer Ch1 (proposed)	Input drive signal level to the klystron from the drive amplifier; sets the klystron gain and operating point.
6	Station Reference Power	Waveform Buffer Ch3	Monitor of the 476 MHz reference signal level distributed to the LLRF system; loss of this signal indicates a reference chain fault.
7	Waveguide Load 1 Forward Power	LLRF9 Unit 2 CH6	Power entering WG Load 1 at port 4 of Magic Tee 1. Rarely exceeds 2 kW.
8	Waveguide Load 1 Reflected Power	Waveform Buffer Ch4	Power reflected from WG Load 1 back into Magic Tee 1; indicates load match quality.
9	Cavity A Forward Power	LLRF9 Unit 1 CH5	RF power traveling into Cavity A from Magic Tee 2; used to monitor power distribution balance and provide reference phase for cavity probe phase measurement.
10	Cavity A Reflected Power	LLRF9 Unit 2 CH0	Power reflected from Cavity A back into the waveguide; elevated reflection indicates cavity detuning or mismatch.
11	Cavity B Forward Power	LLRF9 Unit 1 CH6	RF power traveling into Cavity B from Magic Tee 2; compared with Cavity A forward to verify magic-tee balance and provide reference phase for cavity probe phase measurement.
12	Cavity B Reflected Power	LLRF9 Unit 2 CH1	Power reflected from Cavity B; monitored for detuning and arc detection.
13	Cavity A Probe Signal	LLRF9 Unit 1 CH0	Electric field sampled by Cavity A's internal pickup probe; direct measure of the accelerating voltage amplitude and phase; primary LLRF fast feedback and tuner control input for Cavity A.
14	Cavity B Probe Signal	LLRF9 Unit 1 CH1	Electric field from Cavity B's internal probe; used in the LLRF vector-sum feedback and Cavity B tuner control.
15	Waveguide Load 2 Forward Power	LLRF9 Unit 2 CH7	Power entering WG Load 2 at port 4 of Magic Tee 2; absorbs the sum of equal reflected power from Cavities A and B.
16	Waveguide Load 2 Reflected Power	Waveform Buffer Ch5	Power reflected from WG Load 2; indicates match quality of the load.

#	Signal	Monitored By	Notes
17	Cavity C Forward Power	LLRF9 Unit 1 CH7	RF power traveling into Cavity C from Magic Tee 3.
18	Cavity C Reflected Power	LLRF9 Unit 2 CH2	Power reflected from Cavity C; monitored for detuning and arc detection.
19	Cavity D Forward Power	LLRF9 Unit 1 CH8	RF power traveling into Cavity D from Magic Tee 3.
20	Cavity D Reflected Power	LLRF9 Unit 2 CH3	Power reflected from Cavity D.
21	Cavity C Probe Signal	LLRF9 Unit 1 CH3	Electric field from Cavity C's internal probe; used in the LLRF vector-sum feedback and Cavity C tuner control.
22	Cavity D Probe Signal	LLRF9 Unit 1 CH4	Electric field from Cavity D's internal probe; used in the LLRF vector-sum feedback and Cavity D tuner control.
23	Waveguide Load 3 Forward Power	LLRF9 Unit 2 CH8	Power entering WG Load 3 at port 4 of Magic Tee 3; absorbs the sum of equal reflected power from Cavities C and D.
24	Waveguide Load 3 Reflected Power	Waveform Buffer Ch6	Power reflected from WG Load 3; indicates match quality of the load.

5 Subsystem 1: LLRF Controller

Detailed reference: `llrf/llrf9`

5.1 Legacy System

The legacy LLRF controller is a custom PEP-II analog RF Processor (RFP) module in a VXI chassis. It performs analog I/Q processing at ~+/-90 kHz bandwidth for the fast RF feedback loop. Associated modules include the Clock Module and three IQA (~1 MHz amplitude and phase detector) modules. The VXI chassis also hosts a processor running VxWorks RTOS with SNL (State Notation Language) control programs.

The legacy system processes 24 RF channels through the VXI system and uses analog-domain signal processing for feedback and calibration.

5.2 Upgraded System — Dimtel LLRF9/476

Two Dimtel LLRF9/476 units replace the entire VXI-based LLRF system (four units purchased; two active, two spares).

Hardware per unit:

- 3 × LLRF4.6 boards: each with a Xilinx FPGA, 4 high-speed ADC channels, 2 DAC channels, 3 RF channels
- LO/Interconnect module: divide-and-mix LO synthesis for low phase noise, RF reference distribution, output amplification/filtering, interlock logic
- Linux SBC (mini-ITX): runs the built-in EPICS IOC. Dmitry upgraded it from EPICS Base 3.14 to EPICS 7 in June 2026, so it now matches the SSRL site standard.
- Thermal stabilization: aluminum cold plate with 3 TEC modules under PID control, only available to board1&2
- Power supply: 90-264 VAC auto-ranging
- 3U 19" rack chassis

LO Frequency Plan (LLRF9/476) — nominal frequency 476.3 MHz (current: 476.3051755 MHz):

Signal	Ratio to f_rf	Nominal Frequency (MHz)
Reference (f_rf)	1	476.3052
IF	1/12	39.6921
Local Oscillator	11/12	436.6131
ADC Clock	11/48	109.1533
DAC Clock	11/24	218.3065

Auxiliary I/O per unit:

- 8 slow ADC channels (12-bit, selectable ranges, galvanically isolated)
- 12 slow DAC outputs (AD5644)
- Opto-isolated interlock input (3.3V/5V/24V selectable — confirm exact voltage logic levels from test summaries)
- Interlock output (+4.75V high / <0.1V tripped, 220 Ω)
- Two opto-isolated trigger inputs

5.3 Two-Unit Configuration for SPEAR3

The two units are cabled as follows. Inputs CH0–CH2 belong to LLRF4.7 board 1, CH3–CH5 to board 2 and CH6–CH8 to board 3; the fourth input of every board takes the station reference (RF REF), against which the phases are measured. Only boards 1 and 2 have thermally stabilized output chains. Cavities 1 to 4 are cavities A to D of Section 4.6.

Connector	Unit 1 (master)	Unit 2 (slave)
CH0	Cavity 1 probe	Cavity 1 reflected
CH1	Cavity 2 probe	Cavity 2 reflected
CH2	Klystron forward	Cavity 3 reflected
OUT0	Klystron drive 1	—
OUT1	—	—
CH3	Cavity 3 probe	Cavity 4 reflected
CH4	Cavity 4 probe	Klystron reflected
CH5	Cavity 1 forward	Circulator load forward
OUT2	Klystron drive 2	—
RF REF	Station reference	Station reference
CH6	Cavity 2 forward	Waveguide load 1 forward
CH7	Cavity 3 forward	Waveguide load 2 forward
CH8	Cavity 4 forward	Waveguide load 3 forward
OUT3	—	—

Unit 1 (master) — Field Control & Tuner Loops:

- **Feedback Architecture**: Cavity probes 1 and 2 (board 1) can be combined with proportional and integral control driving OUT0 (klystron drive 1). Probes 3 and 4 (board 2) can be combined with integral control driving OUT2 (klystron drive 2). An external power combiner combines the two outputs to generate a signal containing an integral of all four cavities. The outputs of the two boards are synchronized via phase controls.
- **Tuner loops**: all four cavity probe and forward signals are on this unit, so it runs the four tuner loops on synchronized 10 Hz phase data and drives the Galil DMC-4143 directly through its built-in driver (Section 10.3)
- **Source of the allocation**: this is the vendor's four-cavity configuration, and the master's inputs match the 2021 LLRF9 setup at patch panel B132-11 channel for channel (`llrf/calibrations/b132R11PatchPanel.xlsx`)

Unit 2 (slave) — Monitoring & Interlocks:

- **Reflected power monitoring**: All 4 cavity reflected + klystron reflected for arc/mismatch detection
- **Load monitoring**: circulator load forward and the three waveguide-load forward signals
- **Interlock chain**: Reflected power events → Unit 2 interlock output → Interface Chassis → disables Unit 1 drive
- **No drive output required** from Unit 2

Remaining RF signals: The 6 signals not covered by the two LLRF9 units (Klystron Drive Power, Circulator Load Refl, Station Reference, WG Load 1/2/3 Reflected) are assigned to the Waveform Buffer System, the drive power provisionally (O4) — see Section 11.

5.4 Key Performance Specifications

Parameter	Value
Direct loop delay	270 ns
RF input channels	9 per unit (18 total)
RF input range	+2 dBm full-scale
ADC resolution	12-bit
Setpoint profile points	512
Setpoint step time range	70 μs to 37 ms per step
Waveform samples/channel	16,384
Scalar readback rate	10 Hz
Phase readback resolution	Sub-degree
Interlock timestamp resolution	$\pm$17.4 ns

5.5 Interfaces

- **RF inputs**: 50 Ω SMA from cavity probes, forward couplers, reflected couplers, station reference
- **RF outputs**: Unit 1 OUT0 and OUT2 (klystron drive 1 and 2, the thermally stabilized outputs of boards 1 and 2), combined externally into the drive amplifier; Unit 2 has no drive output
- **Interlock I/O**: LEMO connectors; Unit 1 external interlock input from Interface Chassis, Unit 2 interlock output to Interface Chassis (exact voltage logic levels still to be confirmed from the LLRF9 test summaries **[O1]**)
- **Slow ADC**: DA-15 connector; HVPS monitoring signals, auxiliary sensors
- **Ethernet**: Channel Access for EPICS communication (default ports 5064/5065)
- **Unit 1 ↔ Unit 2**: Interlock daisy-chain (e.g. Unit 2 reflected power trip disables Unit 1 drive)

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6 Subsystem 2: High Voltage Power Supply (HVPS)

Primary source references: Architecture: `hvps/architecture/`

6.1 Power Section (Retained)

Based on the proven PEP-II design architecture from 1997, this legacy system has been adapted for SPEAR3 operational requirements while maintaining the innovative star point controller topology and multi-layer arc protection system. The HVPS converts 12.47 kV RMS 3-phase AC to regulated DC high voltage for the klystron cathode. It is **rated −90 kV at 27 A (2.5 MW)** and typically operates at ≈ **−72 to −75 kV at 19–22 A (≈ 1.4–1.6 MW DC)**, which supports the klystron's ≈ 800 kW typical RF output. The power section, including transformers, thyristor stacks, filter inductors, secondary rectifiers, crowbar, oil system, and all power cabling, is retained unchanged.

Parameter	Value
Maximum output voltage	−90 kV DC
Maximum output current	27 A
Maximum output power	2.5 MW
Nominal operating voltage	−74.4 kV
Nominal operating current	22.0 A
Nominal operating beam current	500 mA

Parameter	Value
Rectifier topology	12-pulse, thyristor phase-controlled (two 6-pulse bridges, B+ and B−)
Number of HVPSs	2 (SPEAR1 and SPEAR2) — see note below

Spare-supply posture. The two supplies are **not** in a warm-standby arrangement. One supply is energised and the other is **off**. Roles are exchanged during scheduled downtimes (typically one in April and one in August); either supply may be the active one. Both are fully operational.

T0 rating. The phase-shifting transformer nameplate is $\mathbf{2\,750}\,\mathrm{kVA}$**,** $\mathbf{12.5}\,\mathrm{kV}$ **at** $\mathbf{127}\,\mathrm{A}$ **RMS**, per NWL's own electrical schematic EI-730-790-00-C0 (NWL #39308). The two nameplate figures are internally consistent:

$$I = \frac{S}{\sqrt{3}\,V} = \frac{2\,750\,\mathrm{kVA}}{\sqrt{3}\times 12.5\,\mathrm{kV}} = 127\,\mathrm{A}.$$

This supersedes the $350\,\mathrm{kVA}$ figure in PS-341-360-01 §1.5 (a decimal error — far too small to pass $2.5\,\mathrm{MW}$) and the rounded "3000 KVA" label on SD-730-790-01-C1.

6.2 Legacy Controller

The legacy controller is housed in a Hoffman NEMA enclosure (34"×42") in Building B118 and contains:

- **PLC**: Allen-Bradley SLC-500 (1747-L532 CPU) with I/O modules (OBSOLETE):

Slot	Module	Function
CPU	AB-1747-L532	Processor
1	AB-1747-DCM	Data Communications (Scanner)
2	AB-1746-IO8	8-pt Digital I/O — Ross switch coil (OUT3)
3	AB-1746-THERMC	Thermocouple inputs (SCR/air/transformer temps)
5	AB-1746-OX8	8-pt Relay Output — Contactor enable K4 (OUT2), Contactor On/Off (OUT1)
6	AB-1746-IB16	16 DC Input — PPS 1 (IN14), PPS 2 (IN15), Oil Level (IN8), Manual GRN SW (IN9)
7	AB-1746-IV16	16 DC Input (various permits)
8	AB-1746-NIO4V	4-ch Analog I/O — voltage setpoint output to Enerpro via regulator (N7:10 → OUT0)
9	AB-1746-NI4	4-ch Analog Input — Danfysik HVPS current (IN3)

Source: `hvps/documentation/plc/plcNotesR1.docx` (N7:10/N7:11 analysis, slot assignments)

- **Enerpro Firing Board**: FCOG6100 + FCOAUX60 (30° delayed triggering daughter board) — Current SCR gate driver
- **Regulator Card** (SD-237-230-14-C1): SLAC-designed analog voltage regulation loop with INA117 (difference amp), INA114 (instrumentation amp), OP77 (op-amp, 600 kHz GBW), BUF634 (high-current buffer), 4N32 optocoupler, and VTL5C opto-controlled variable resistor (OBSOLETE component)
- **Power Supplies**: SOLA ±15V/+5V/24V, Kepko 120V (×2), Kepko 5V/20A, Kepko 240V
- **PS Monitor Board** (SD-730-793-12): Monitors power supply health

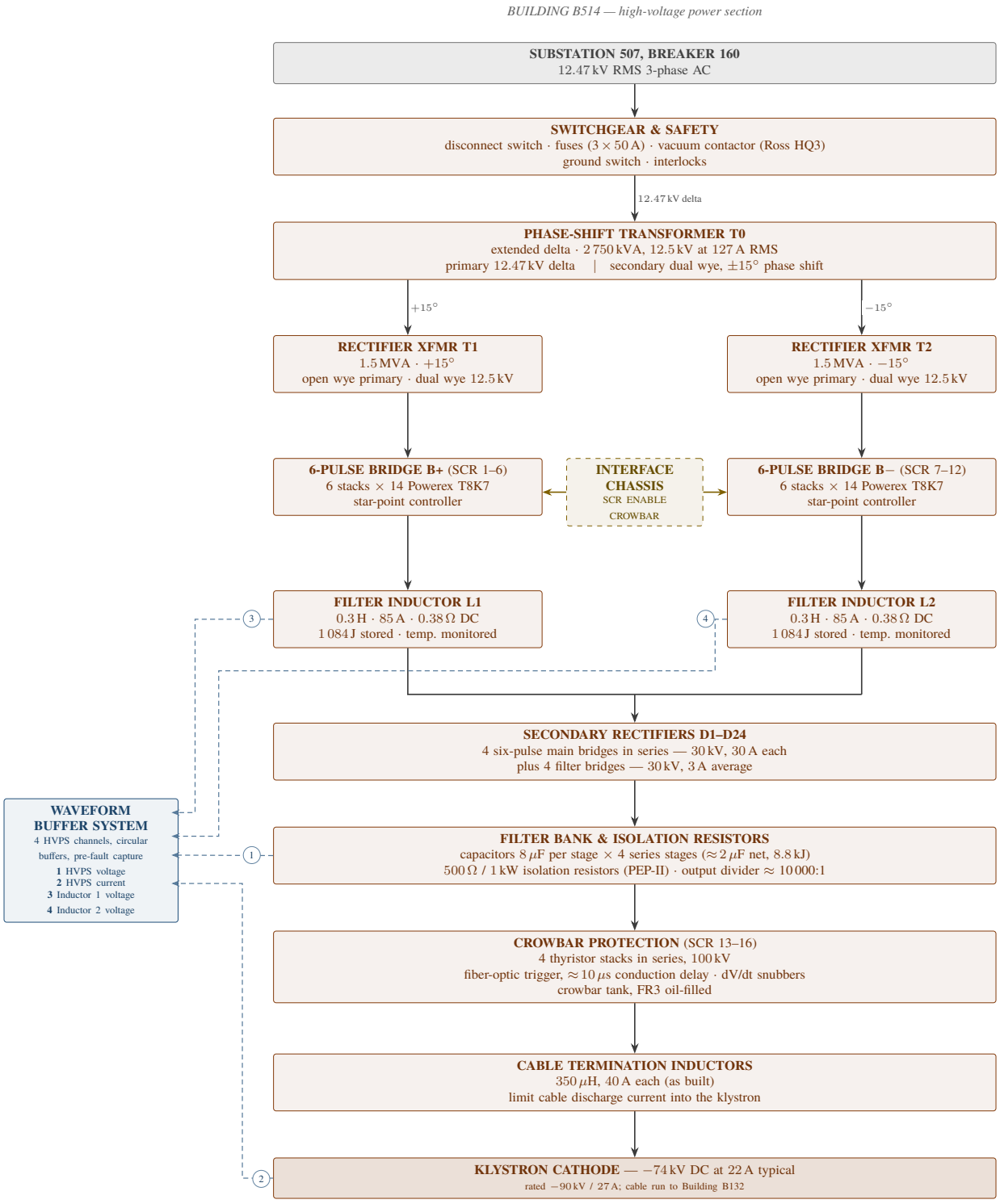


Figure 3: HVPS power chain in Building B514, from the incoming substation feeder to the klystron cathode. The 15° phase offset between the two rectifier transformers is what turns two 6-pulse bridges into a single 12-pulse supply. Dashed blue lines are the four Waveform Buffer monitoring taps, numbered as in §11.2; they are the only non-power signals shown. Component values are the as-built ones verified against the original drawings — T0 is 2 750 kVA, the filter capacitors are 8 μF *per stage* across four series stages, the output divider is ≈ 10 000:1, and the cable termination inductors are 350 μH.

- **Terminal Strips**: TS-5 (contactor, 15 terminals), TS-6 (grounding tank, 21 terminals), TS-3 (PPS LEDs), TS-7 (power distribution)
- **PPS Connector**: Originally Burndy circular 8-pin; possibly replaced with Souriau Trim Trio equivalent (GOB1208PNE) — exact current part number requires field verification [O2]

The SLC-500 PLC executes ladder logic for voltage regulation, contactor management, temperature monitoring, and — critically — PPS safety chain control:

- **PPS 1**: Controls K4 enable path for 12kV contactor; also has hardware fail-safe wired directly to OX8 relay input (bypasses PLC)
- **PPS 2**: Controls Ross grounding switch coil via Slot-2 IO8 OUT3 (Rung 0016) (Ross relay shorts out HV input to klystron.)
- **Hardware fail-safe**: K4 relay input side wired directly to PPS 1 signal (24 VDC), preventing PLC failure from keeping K4 energized without PPS enable

> **Sources**: `hvps/architecture/designNotes/HoffmanBoxPPSWiring.docx`, `hvps/architecture/designNotes/RFSystemMPSRequirements.docx`
>
> **Note on PPS chain status**:
>
> - **Current system**: only *overall* PPS status is reported. One chain is reported directly and the other only indirectly, so the two cannot be distinguished at the RF station. Determining which chain caused a given event means going to the SPEAR Event Log (`https://www-ssrl.slac.stanford.edu/spear/eventlog`) and inferring it after the fact.
> - **Upgraded system**: each PPS chain gets its own dedicated status signal, taken from an auxiliary contact on a relay energised by that chain's PPS signal. Both chains are then monitored and logged directly, and no inference from the event log is needed.

6.3 Upgraded Controller

The HVPS controller upgrade builds a complete new Hoffman box for each HVPS unit, which retires the existing box with its SLC-500, firing and regulator boards, trigger interconnect boards, power supplies and local control panel. The power section and the field cables to it, to the contactor controller and to the grounding tank are retained.

New hardware:

- **CompactLogix PLC**: Replaces SLC-500; handles voltage regulation, temperature monitoring, fault management, and EPICS communication
- **Enerpro FCOG1200 Board**: Upgraded SCR gate driver with modified RN4 resistors for SPEAR monitor winding impedance (2 MΩ)
- **Redesigned Analog Regulator**: Replaces obsolete regulator card components
- **Fiber Optic Interfaces**: Bidirectional communication with Interface Chassis:
  - STATUS: HVPS controller → HFBR-1412 transmitter → Interface Chassis (indicates control supply presence + no crowbar fired)
  - SCR ENABLE: Interface Chassis → HFBR-2412 receiver → HVPS controller (permit signal, active-high for fail-safe)
  - CROWBAR: Interface Chassis → HFBR-2412 receiver → HVPS controller (crowbar fire command)
  - Normal operation: All signals illuminated/active; signal loss = fault condition

Key change: The CompactLogix PLC is removed from the PPS safety chain. PPS functions (K4 relay, Ross switch control) are handled by a separate dedicated PPS interface box, not the Interface Chassis. The Interface Chassis handles only non-PPS interlocks. The PLC handles only non-safety functions: voltage regulation, temperature monitoring, and EPICS interface.

6.4 HVPS Controller Interface

With EPICS:

PV Category	Example PV	Update Rate
Setpoint	`SRF1:HVPS:VOLT:CTRL.VAL`	$\leq$1 Hz (EPICS HVPS loop)
Readback	`SRF1:HVPS:VOLT:RBCK`	~1 Hz
Status	`SRF1:HVPSSUPPLY:READY:STAT`	~1 Hz
Control	`SRF1:HVPSCONTACT:CLOSE:CTRL`	On demand
Interlock	`SRF1:HVPSOFF:SUMY:STAT`	~1 Hz

All HVPS control stays in EPICS, in the IOC B132-IOCRF, with no Python layer. The upgraded controller keeps the current PV names of the legacy HVPS database (`spear-rf-code-legacy/rfApp/Db/rf_hvps.db,v`), so the screens, alarms and archiver that use them need no change.

Legacy HVPS Status PVs (for reference and troubleshooting):

Status Parameter	Function	Source
Contactor Closed	K4 relay status / Aux relay status	Slot-6 inputs
Over Voltage	Regulator card latch indication	SD-237-230-14-C1
Klystron Arc	Grounding tank + LHS trigger board detection	BNC-12 input
Transformer Arc	HVPS + RHS trigger board detection	BNC-0 input
Crowbar Status	Thyristor pulse detection	Enerpro board
PPS Chain Status	PPS 1 and PPS 2 availability (upgraded: both chains via relay aux contact)	Slot-6 IN14/IN15
SCR 1/2 Firing	Left/right side thyristor trigger status	Enerpro monitoring
AC Current	Input current measurement	Transformer monitoring
Temperature/Oil	SCR/air/transformer temperatures, oil level	Slot-3 thermocouples
12 kV Availability	Supply status/readiness	Switchgear feedback
Fast Inhibit/Slow Start	Enerpro protection states	Gate driver board

Source: `hvps/architecture/designNotes/rfedmHvpsLabelsPvs.docx` (RF expert panel configuration)

With Interface Chassis:

- **Interface Chassis** (fiber optic): SCR ENABLE (in), CROWBAR inhibit (in), STATUS (out)

Crowbar Triggering Sources (five independent sources for defense-in-depth protection):

1. **SCR ENABLE** (fiber-optic): Interface Chassis $\rightarrow$ HVPS controller (loss of signal disables SCR triggers)
2. **TRANSFORMER ARC TRIGGER** (BNC-0): Electronic signal from Hoffman box (AC current measurement or Stangenes transformer arc detection)
3. **KLYSTRON CROWBAR** (fiber-optic): LLRF/Interface Chassis $\rightarrow$ HVPS controller (klystron protection command)

4. **KLYSTRON ARC TRIGGER** (BNC-12): Electronic signal from Hoffman box (termination tank shunt sensing excessive current)
5. **PLC FORCE CROWBAR** (Slot-5 OUT3): Active-low signal from SLC-500, monitored on BNC connector

> **Protection philosophy**: Any single source can disable SCR triggers or fire crowbar; redundancy ensures protection even with single-point failures. Filter inductor stored energy is safely discharged through Enerpro logic and secondary rectifiers. Source: `hvps/architecture/designNotes/RFSystemMPSRequirements.docx`

- **EPICS Coordinator** (Ethernet): Voltage setpoint, readbacks, fault status
- **Waveform Buffer System**: HVPS voltage, current, inductor voltages monitored on 4 dedicated channels. HVPS voltage and current sent back to B132 as analog signals for RF MPS.
- **Switchgear**: Existing field cables to vacuum contactor controller and grounding tank

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7 Subsystem 3: RF Machine Protection System (RF MPS)

> **Scope clarification**: The RF MPS is the klystron/RF station equipment protection PLC. It is distinct from the facility-wide **SPEAR MPS**, which is an external system that provides a permit input signal to the RF MPS via the Interface Chassis.

7.1 Purpose and Protection Philosophy

The RF Machine Protection System (RF MPS) is the subsystem-level protection controller for the SPEAR3 RF station. Its purpose is to aggregate fault status from all RF subsystems and external safety systems (including the SPEAR MPS permit), manage the overall klystron/RF permit, and coordinate fault recovery.

The protection philosophy, as defined in the RF system MPS requirements (`hvps/architecture/designNotes/RFSystemMPSRequirements.docx`), is:

- Protect high-power elements (HVPS controller protects the HVPS; the RF MPS system protects the klystron and cavities) from damage by eliminating stored energy and disabling upstream power sources when faults occur.
- Redundant protection — multiple independent mechanisms protect each element (e.g., fiber-optic SCR disable, crowbar firing, contactor opening).
- Fail-safe design — all permits are active-high so that a broken cable or lost signal removes the permit.
- Graceful shutdown — when upstream elements trip, downstream elements are notified so they can shut down in an orderly fashion.

The RF MPS does **not** directly control the LLRF9 or HVPS hardware. Instead, it provides permit and control signals to the **Interface Chassis** (Section 8), which performs the fast hardware AND-logic and drives the actual interlock outputs to the LLRF9 and HVPS controller. This separation ensures that hardware interlock response times remain at the microsecond scale (Interface Chassis), while the RF MPS operates at PLC scan rates (~milliseconds) for fault aggregation, logging, and coordination.

7.2 Legacy System

The legacy RF MPS uses an **Allen-Bradley PLC-5** processor with **1771-series I/O modules**. It aggregates interlock signals from the RF system (via distributed analog modules and direct wiring) and the HVPS, and

manages the RF permit chain. The SPEAR MPS beam permit and orbit interlock are **not** direct inputs to the legacy RF MPS PLC — they wire to the back connector of VXI Slot 5 (MPS Shutoff module) where they participate in the VXI-backplane RF permit logic. See Doc I §4.2.

In the legacy system, interlock coordination was distributed across analog modules, PLC I/O, and direct point-to-point wiring with no central coordination point. The PLC-5 communicated with the EPICS control system via a 1771-DCM scanner module.

> **Legacy RF MPS wiring diagrams**: 33 sheets in `llrf/documentation/mpsWiringDiagrams/` (drawings wd3403300200 through wd3403303400).

7.3 Upgraded System — ControlLogix 1756

The RF MPS PLC is upgraded from PLC-5 to **Allen-Bradley ControlLogix 1756** platform using a Rockwell Automation conversion kit. The ControlLogix platform provides modern Ethernet/IP communication, faster scan times, and long-term vendor support.

Status: Hardware assembled, software written, tested on SPEAR3 without RF power. Ready for EPICS IOC development and system integration.

7.4 RF MPS Outputs to Interface Chassis

In the upgraded system, the RF MPS communicates with the Interface Chassis via digital signals. These are the **only** direct hardware outputs from the RF MPS PLC to the interlock system:

Output Signal	Type	Description
RF MPS Summary Permit	Digital	Global RF permit from the RF MPS. One of several AND-gate inputs in the Interface Chassis. When removed, the Interface Chassis removes LLRF9 Enable and HVPS SCR ENABLE.
RF MPS Heartbeat	Digital	Watchdog signal. If the Interface Chassis stops receiving the heartbeat (indicating RF MPS PLC failure or communication loss), it removes all output permits.
RF MPS Reset	Digital	Clears all latched faults in the Interface Chassis first-fault register simultaneously. Required to restore system permits after a fault.

Important: The RF MPS does **not** directly drive the LLRF9 Enable, HVPS SCR ENABLE, or HVPS KLYSTRON CROWBAR signals. Those are Interface Chassis outputs, driven by the AND-logic of all input permits (see Section 8). The RF MPS contributes its Summary Permit as one input to that AND gate.

7.5 RF MPS Inputs from Interface Chassis

The RF MPS receives comprehensive status from the Interface Chassis via a multi-conductor digital cable:

Input Signal	Description
All input permit states	Status of every Interface Chassis input (LLRF9 Status, HVPS STATUS, SPEAR MPS permit, Orbit Interlock, Arc Detection, Waveform Buffer, expansion ports)

Input Signal	Description
All output permit states	Status of every Interface Chassis output (LLRF9 Enable, HVPS SCR ENABLE, HVPS CROWBAR, beam abort to the SPEAR MPS)
First-fault register	Identifies the initiating fault source when multiple faults cascade. Hardware-latched at the moment of initial fault detection.

This information allows the RF MPS to:

- Determine which subsystem caused a trip
- Log complete fault histories with timestamps
- Report detailed status to the EPICS coordinator for operator diagnostics

7.6 RF MPS Role in the Protection Chain

The RF system implements a layered protection architecture. The RF MPS operates at **Layer 3** (PLC scan rate, ~milliseconds):

Protection Layer	Subsystem	Response Time	Function
Layer 1	LLRF9 FPGA	few μs	RF overvoltage interlocks, baseband window comparators, DAC zeroing, RF switch
Layer 2	Interface Chassis	few μs	Hardware AND-logic, first-fault latching, HVPS fiber-optic control, LLRF9 enable
Layer 3	RF MPS PLC	~ms (PLC scan)	Fault aggregation, permit management, reset coordination, event logging
Layer 4	EPICS coordinator	~1 Hz	State machine, supervisory control, operator interface, auto-recovery

The RF MPS functions include:

- Aggregating fault status from all RF station subsystems (via Interface Chassis digital status lines)
- Receiving the SPEAR MPS permit as one of the AND-gate inputs via the Interface Chassis
- Providing the RF MPS Summary Permit to the Interface Chassis AND-gate
- Sending the heartbeat watchdog signal to the Interface Chassis
- Issuing the reset signal to the Interface Chassis to clear latched faults after a trip
- Logging fault events with timestamps for post-mortem analysis
- Providing redundant collector power calculation (from Waveform Buffer System data sent to RF MPS for independent verification)
- Reporting complete system status to the EPICS/MATLAB coordinator

7.7 EPICS Interface

The RF MPS ControlLogix PLC will provide an EPICS interface for operator monitoring and diagnostics. The PV namespace is `SRF1:MPS:`. Key PVs include:

PV	Type	Description
SRF1:MPS:PERMIT	BINARY	Overall MPS permit status
SRF1:MPS:FAULTS:ACTIVE	MULTI-BIT	Currently active fault sources
SRF1:MPS:FAULTS:FIRST	ENUM	First-fault identification (from Interface Chassis)
SRF1:MPS:FAULTS:COUNT	INTEGER	Accumulated fault event count

The EPICS coordinator reads these PVs to:

- Determine whether the MPS permit is active before allowing state transitions (e.g., OFF → STANDBY → ON)
- Force a transition to OFF state if MPS permit is lost during operation
- Display fault summaries and first-fault information to operators

7.8 Interfaces Summary

Interface Partner	Signal Direction	Signals	Medium
Interface Chassis	RF MPS → IC	Summary Permit, Heartbeat, Reset	Digital (multi-conductor cable)
Interface Chassis	IC → MPS	All input/output states, first-fault register	Digital (multi-conductor cable)
EPICS Coordinator	Bidirectional	Permit status, fault history, reset commands	EPICS Channel Access (Ethernet)

Note: External safety systems (SPEAR MPS permit, orbit interlock) connect to the **Interface Chassis** directly, not to the MPS PLC. The MPS receives their status indirectly via the Interface Chassis digital status lines. This design ensures that external safety permits participate in the fast hardware AND-logic (few μs) within the Interface Chassis, rather than being limited to PLC scan rates.

7.9 Source Documents

File	Description
hvps/architecture/designNotes/RF SystemMPSRequirements.docx	RF system machine protection requirements and protection philosophy (J. Sebek)
hvps/architecture/designNotes/ interfacesBetweenRFSystemControllers. docx	Interface Chassis design — defines the MPS-to-Interface Chassis signal interface
llrf/documentation/mpsWiringD iagrams/	33 legacy MPS wiring diagram sheets (wd3403300200–wd3403303400)
pps/HoffmanBoxPPSWiring.docx	System-level architecture showing MPS in the upgraded system context

File	Description
`llrf/llrf9/iGp/dl_llrf/interlock_summary.edl`	LLRF9 interlock summary display (EDM) — shows LLRF9-side interlock PVs that feed into the protection chain

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8 Subsystem 4: Interface Chassis

Detailed reference: `llrf/architecture/llrfInterfaceChassis.docx`

8.1 Purpose

The Interface Chassis is a completely new subsystem that serves as the central hub for all hardware interlock coordination. It did not exist in the legacy system, where interlocks were distributed across analog modules, PLC I/O, and direct wiring with no central coordination point.

It absorbs the functions of three legacy units that are removed outright (Section 2.3): the **Fast Interlock Chassis 340-308** (fast trip decision, arc inputs, reflected-power comparators), the **AIM** in VXI Slot 12 (per-channel arc latching and `RF_FAULT` assertion — the fault-history buffer function goes to the Waveform Buffer instead), and the **RF Local Display Panel 340-310** (the logic that forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort to the SPEAR MPS, the end of the three HVPS fibers, and local control and indication). Consolidating all three into one chassis is what makes single-point first-fault detection possible; in the legacy system the trip decision, its digital record and the operator indication lived in three different boxes. The panel's I/O is documented in its drawings and in J. Sebek's fiber note, but the equations of its programmable logic are not, and the endpoint of its beam-abort output in the SPEAR MPS is not on any RF drawing. **[O3]**

Key capabilities:

- **First-fault detection** — hardware latching identifies the initiating fault in cascade scenarios
- **Microsecond-scale response** — combinational logic with no processor in the critical path
- **Electrical isolation** — all external signals isolated via optocouplers and/or other galvanic isolation and fiber-optic transceivers
- **Fault latching** — all inputs latch when faulted until external MPS reset
- **Status reporting** — comprehensive digital reporting to MPS PLC

8.2 Interface Summary

Primary inputs: LLRF9 status, HVPS fiber-optic status, MPS permits/heartbeat/reset, SPEAR MPS, orbit interlock, arc detection, power monitoring, expansion ports

Primary outputs: LLRF9 enable, HVPS SCR ENABLE (fiber), HVPS CROWBAR (fiber), beam abort to the SPEAR MPS, fault status reporting to MPS

Beam abort: the legacy station tells the SPEAR MPS that it is not ready for beam through the beam-abort output of the RF Local Display Panel (rear BNC J6). The panel combines the beam abort that the state machine forces through the AIM whenever it turns the station off or enters TUNE, the HVPS STATUS fiber and the main and bucking solenoid faults, and latches the result until reset. The Interface Chassis keeps this output with the same meaning.

Critical design consideration: The Interface Chassis creates a bidirectional feedback loop between LLRF9 and HVPS that requires careful recovery sequencing logic.

8.3 Transition Interfaces

The Interface Chassis goes in early: it replaces the Fast Interlock Chassis and the RF Local Display Panel while the AIM, the RF MPS PLC-5 and the legacy HVPS controllers are still in service (integration I2, Section 19.2). Until those units are retired it must also present the interfaces of the two units it replaces:

- **AIM cable J15**: accepted unchanged. The chassis receives the AIM's six commands (station online, filament and focus, HVPS on, beam abort, filament-timeout bypass, fault reset) at their present levels, drives the AB summary fault (pin 10) and the AB heartbeat (pin 23) into the AIM's Schmitt-trigger inputs, and terminates the other AIM lines as the Fast Interlock Chassis does.
- **RF MPS cables J2 and J3 at cross-connect X530**: accepted unchanged, with every line reproduced — the requests, the HV-detected, RF-detected, local and online indications, the filament and coil-supply controls, the 15 area summaries, the AB summary and the AB heartbeat.
- **HVPS voltage and current**: buffered into the PLC-5's analog inputs with unchanged scaling, because the PLC-5 computes the collector power from them.
- **Klystron forward power**: the output of the Fast Interlock Chassis's forward-power detector (Section 4.5) reproduced with unchanged scaling while the PLC-5 uses it.
- **Panel logic**: the equations, latching and reset of the panel's GAL22V10, which forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort from the HVPS-on request, the VXI and HVPS beam aborts, the main and bucking solenoid faults, the waveguide-pressure and collector summaries and the online key switch; every deliberate difference is documented and reviewed.
- **Operator controls**: a status display and the station online/offline and local/remote key switches, with all states also in EPICS.

The end-state inputs (LLRF9 status and enable, SPEAR MPS and orbit permits, arc detection, power monitoring, expansion ports) are built in from the start and masked, each with a bypass or simulate jumper, until the integration that uses them. A command path from the Interface Chassis IOC replaces the AIM cable J15 when the AIM is retired with the VXI crate.

8.4 Implementation Status

Shared physical chassis: the Interface Chassis and the Waveform Buffer System (Section 11) are **one small physical chassis carrying two separate PCBs** — one board per subsystem. They remain functionally distinct subsystems with distinct interfaces, and this report continues to describe them separately, but they are procured, fabricated and installed as a single unit.

Item	Status
Interface specification	In progress (J. Sebek, `llrf/architecture/llrfInterfaceChassis.docx`)
Chassis design	Shared with the Waveform Buffer board; being raised from 2U to **3U**
Prototype	Built
Panels	Front and back panels to be modified for the 3U chassis
Bench test	Planned October–November 2026

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9 Subsystem 5: Personnel Protection System (PPS) Interface

Detailed reference: `pps/`

9.1 PPS Overview

The PPS is the radiation and electrical safety interlock system managed by SLAC's radiation protection group. It controls two points in the HVPS power chain:

- **Chain 1**: Vacuum contactor (removes 12.47 kV input power to HVPS)
- **Chain 2**: Ross grounding switch (shorts the HVPS output to ground)

The PPS interface uses a GOB1208PNE (Burndy/Souriau Trim Trio) 8-pin circular connector mounted in a locked box on the Hoffman enclosure.

9.2 Legacy PPS Design (Problems)

In the legacy system, PPS signals are routed through the Hoffman Box:

- **PPS wiring exposure** — PPS wires terminate on terminal strips (TS-5, TS-6) inside the HVPS controller, co-located with non-PPS wiring
- **PLC in safety chain** — The Ross grounding switch is controlled by the SLC-500 PLC (Slot-2 IO8 OUT3, 120 VAC). If the PLC fails in a stuck-on state, the Ross switch coil stays energized (unsafe)
- **K4 relay via PLC** — Although the K4 relay coil uses PPS 1 voltage for its power rail (providing a hardware fail-safe), the enable path runs through PLC OX8 OUT2

9.3 Upgraded PPS Design

The upgraded design adopts the standard PPS interface solution already approved and in use by the PPS group for the 6575 modulator and gallery systems. It uses a dedicated PPS interface box.

PPS Interface Box Characteristics (per Ben Morris, March 5, 2026):

- Small Bud enclosure ($\approx 6 \times 5 \times 4$ inches)
- Contains 4 relays, status LEDs (Permit A/B granted & enabled), and inhibit push-buttons
- Single board with one connector that the PPS group can lock with their collar
- Provides two independent permit channels + two readback channels (dry contacts)
- Completely isolated from the HVPS internals
- Labeled "PPS Interface – RSWCF required to open" (standard PPS practice)

PPS Signal Flow:

- **Chain 1 (Contactor)**: PPS Enable 1 $\rightarrow$ PPS Interface Box $\rightarrow$ HVPS contactor relay direct drive $\rightarrow$ MX $\rightarrow$ L1 holding coil. S5 NC auxiliary contact readback via PPS Interface Box to PPS.
- **Chain 2 (Ross Switch)**: PPS Enable 2 $\rightarrow$ PPS Interface Box $\rightarrow$ Ross switch direct drive. Ross NC auxiliary contact readback via PPS Interface Box to PPS.

Additional Safety Relays: Because the inhibit and safety-discharge wiring cannot be physically separated inside the existing HVPS and termination tank, a second relay is added in series with the inhibit/safety-discharge lines (one relay per channel). These relays are labeled "PPS Controlled – RSWCF required before work" with PPS-provided cable tags.

Key improvements:

- No PLC dependency for safety functions
- Uses proven PPS-approved design (identical to gallery systems)
- Electrical isolation of all PPS signals
- PPS wiring completely isolated from non-PPS equipment
- Clear visual feedback and independent inhibit capability
- Meets all current PPS requirements (two independent channels, lockable interface, visible verification)
- Eliminates the "unique RF situation" that the PPS group dislikes

Source: PPS meeting note, 5 March 2026

9.4 PPS Regulatory Considerations

Any modification to PPS wiring, control logic, or readback paths requires:

- Review and approval by the SLAC Accelerator Directorate Safety Office (ADSO)
- Formal change control documentation
- Radiation safety verification testing
- Possibly a radiation safety work control form

This adds significant administrative scope beyond the engineering work and requires early engagement with the PPS/protection group.

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10 Subsystem 6: Tuner Control System

10.1 Physical Mechanism

Each of the 4 RF cavities has a mechanical tuner that adjusts the cavity resonant frequency by moving a plunger into or out of the cavity volume. The tuner is driven by a stepper motor (Superior Electric M093-FC11 or equivalent) through a gear reduction mechanism.

Tuner parameters (from legacy documentation):

- Gear ratio: 80:1 (worm gear)
- Lead screw pitch: 10 turns/inch
- Microstep resolution: 0.002–0.003 mm per microstep (legacy)
- Linear potentiometer: Provides absolute position indication (not used in closed-loop control)
- Travel range: ~12 mm total mechanical range
- ON home positions: ~10.1–10.7 mm (cavity-dependent)
- Park positions: defined in the legacy software; the PARK state is not used at SPEAR3

10.2 Legacy Controller

- **Motor controller**: Allen-Bradley 1746-HSTP1 (high-speed stepper module, OBSOLETE)
- **Motor driver**: Superior Electric SS2000MD4-M Slo-Syn PWM driver (OBSOLETE)
- **Control software**: SNL `rf_tuner_loop.st,v` — phase-based feedback with home reset, motion monitoring, and load angle offset computation
- **Phase source**: Legacy analog RFP module

10.3 Upgraded Controller

The tuner motor controller is replaced by the **Galil DMC-4143** (4-axis). LLRF9 Unit 1 runs the four tuner loops and drives the Galil directly through its built-in Galil driver, so no separate motion-control IOC is needed. The Galil has been built and tested on the SPEAR3 tuner motors under an EPICS control by M. Dunning; that control is not carried forward.

Upgraded tuner control loop:

- LLRF9 Unit 1 compares each cavity's probe phase with its forward phase (10 Hz, synchronized across its boards, sub-degree resolution)
- Its built-in PID tuner loop computes the motor command, with a deadband on the load-angle error and a minimum forward power
- The LLRF9 sends the command to the Galil over Ethernet and reads back the limit switches and an analog input; the tuner motors have no encoders

- The Galil drives the stepper motors

LLRF9 tuner support: The LLRF9 includes built-in tuner control PVs (per cavity):

- `LLRF:TUNER:Cn:GAIN_P` — proportional gain
- `LLRF:TUNER:Cn:GAIN_I` — integral gain
- `LLRF:TUNER:Cn:OFFSET` — detuning phase offset
- `LLRF:TUNER:Cn:CLOSE` — loop state (open/closed)
- `LLRF:TUNER:Cn:MINFWD` — minimum forward power threshold

10.4 Load Angle Offset Loop

The load angle offset loop is a supervisory function that balances gap voltage across all 4 cavities by adjusting the individual tuner phase setpoints. In the legacy system this was embedded in `rf_tuner_loop.st,v`; in the upgrade it becomes a separate Python module (e.g., `load_angle_controller.py`) that:

- Reads all 4 cavity probe amplitudes from LLRF9
- Computes amplitude imbalance
- Adjusts individual tuner phase offset PVs to redistribute power

10.5 Tuner Interfaces

- **LLRF9 Unit 1**: Measures the four cavity probe and forward phases at 10 Hz (probes on CH0, CH1, CH3 and CH4; forward signals on CH5 to CH8) and runs the four tuner loops
- **Galil DMC-4143**: Commanded by LLRF9 Unit 1 over Ethernet through the LLRF9's built-in driver
- **Stepper Motors**: Power and limit-switch cables to tunnel (existing field cabling)
- **EPICS Coordinator**: Opens and closes the LLRF9 tuner loops with the station state and sets the load-angle offsets

10.6 Risk Note

The tuner motor controller is identified as the hardest-to-prove subsystem. Previous attempts with candidate controllers have had reliability issues. The mitigation plan is to test the chosen controller on the SPEAR3 booster tuners first before committing to storage ring installation. The Galil controller has been selected for this purpose due to its proven track record and compatibility with the other facilities. The controller has been tested with SPEAR in Aug 2026 under an EPICS control by M. Dunning, successfully driving the tuner motors of all four cavities simultaneously, which gives confidence for its deployment. In service LLRF9 Unit 1 drives it directly, so that EPICS control is not carried forward.

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11 Subsystem 7: Waveform Buffer System

Detailed reference: `llrf/architecture/WaveformBuffersforLLRFUpgrade.docx`

Status: complete (April 2026). The Waveform Buffer is the first upgrade subsystem finished. It shares **one small physical chassis with the Interface Chassis** (Section 8), on **two separate PCBs** — one board per subsystem. The two remain functionally independent and are described separately throughout this report, but they are fabricated and installed as a single unit, and the Waveform Buffer board is therefore already in hand while the Interface Chassis board is also fabricated. The prototype chassis is done and assembled as a single unit.

11.1 Purpose

The Waveform Buffer System is a new signal conditioning and monitoring chassis that extends the LLRF9's RF monitoring capabilities and adds dedicated HVPS signal monitoring. It serves three functions:

- **Extended RF signal monitoring** — 8 RF channels with circular waveform buffers for pre-fault capture; monitors the 6 RF signals not assigned to the two LLRF9 units
- **HVPS signal monitoring** — 4 channels (voltage, current, 2 inductor voltages) with circular buffers
- **Enhanced klystron collector power protection** — computes DC power minus RF power (vs. legacy forward power proxy) and triggers a hardware trip if collector power exceeds the limit

11.2 Channel Configuration

RF Channels (8):

The two LLRF9 units together cover 18 of the 24 monitored RF signals (see Section 5.3). The 6 remaining signals are routed to the Waveform Buffer RF inputs:

Channel	Signal #	Signal	Purpose
1	5	Klystron Drive Power (proposed)	Drive level into the klystron; the drive power is on neither LLRF9 unit (O4)
2	4	Circulator Load Reflected Power	Indicates circulator load match quality
3	6	Station Reference Power	Monitors 476 MHz reference level; loss of reference triggers fault. Note: Dmitry's LLRF9 module may also measure this.
4	8	Waveguide Load 1 Reflected Power	Monitors WG Load 1 match quality
5	16	Waveguide Load 2 Reflected Power	Monitors WG Load 2 match quality
6	24	Waveguide Load 3 Reflected Power	Monitors WG Load 3 match quality
7	—	(Spare — consider klystron forward power coupled version for collector power calculation)	Reserved [O4]
8	—	(Spare)	Reserved for future use

HVPS Channels (4):

Channel	Signal	Conditioning	Purpose
1	HVPS Voltage	Voltage divider	Regulation monitoring
2	HVPS Current	Current transformer	Load monitoring
3	Inductor 1 Voltage	Voltage divider	Firing circuit health
4	Inductor 2 Voltage	Voltage divider	Firing circuit health

11.3 Key Features

- **Circular buffers**: Continuous acquisition at kHz rate; freeze on fault trigger for pre/post-fault capture (~100 ms pre-fault data)

- **Analog comparator trips**: Hardware-based threshold detection on each channel; comparator outputs feed Interface Chassis
  Enhanced collector power protection algorithm (improvement over legacy forward power proxy):

$$P_{\text{DC}} = V_{\text{HVPS}} \times I_{\text{HVPS}}$$
$$P_{\text{RF}} = P_{\text{klystron forward}}$$
$$P_{\text{collector}} = P_{\text{DC}} - P_{\text{RF}}$$

A hardware comparator trips when $P_{\text{collector}}$ exceeds the configured collector limit.

This provides direct collector power calculation vs. the legacy system's forward power proxy method.

- **EPICS interface**: Waveform readout, threshold configuration, trip status

11.4 Interfaces

- **RF signal inputs**: RF detectors on cavity forward/reflected couplers (signal conditioned)
- **HVPS signal inputs**: Voltage dividers and current transformers from HVPS
- **Interface Chassis**: Comparator trip outputs (digital) feed into Interface Chassis permit logic
- **EPICS Coordinator**: Waveform readout, configuration, collector power trend monitoring

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12 Subsystem 8: Arc Detection System

12.1 Purpose

The arc detection system is a new subsystem that provides optical monitoring of the RF cavity windows, the circulator and the klystron output window for electrical arcing. The legacy capability — fiber-optic arc inputs on the Fast Interlock Chassis 340-308, latched and reported to the VXI backplane by the AIM in Slot 12 — never performed as intended, and it is not used now (Section 2.3).

12.2 Technology — AFT microwave ARC4

Vendor and architecture change — September 2026. Orignally **Microstep-MIS** waveguide arc detectors are specified for the upgrade. That selection has been **abandoned**: the vendor did not respond to repeated technical enquiries from SSRL (R. Lion, K. Vedyanina), and its sensors would in any case have required custom mechanical adapters and a viewport redesign at every cavity. The selected system is the **AFT microwave GmbH (Germany) ARC4**. Its fiber inputs mate directly with the arc-detection fibers already pulled through the tunnel to the four cavities, the circulator and the klystron output window, so **no mechanical adapter and no viewport redesign are required**.

The ARC4 detects the light flash produced by an electrical arc in a waveguide, cavity window or circulator, and converts it into a hardware trip with no processor in the path. It is procured as a **complete vendor system chassis** — not as a set of individual sensors and receivers to be integrated by SSRL.

12.3 System Architecture

The ARC4 chassis is self-contained. SSRL buys and integrates it; SSRL does not design it. It contains:

- Its own **power supply** and **backplane**

- **2 system modules** — these aggregate the state of every detector module across the backplane and present the arc detection system's outputs to the outside world
- **1 test module** — provides the self-test / injected-stimulus path that exercises the chain end to end without a real arc
- **Individual ARC detector modules** — each accepts one optical fiber on the front panel and drives the backplane. **The detector modules have no external outputs of their own.**

This gives the rest of the system a single clean boundary: **the SSRL Interface / Waveform Chassis connects to the 2 AFT system modules and to the test module, and to nothing else inside the ARC4**. No SSRL wire lands on an individual detector module.

12.4 Installation and Channel Assignment

No mechanical work is required. The arc-detection fibers were pulled to the cavities, the circulator and the klystron output window when the tunnel was cabled, and the ARC4 detector modules accept those fibers directly. The custom CF-flange adapters and viewport modifications that the Microstep-MIS option would have needed are out of scope; the existing window viewports (Section 4.3) are used as installed.

There are **10 arc channels** in the current plant configuration:

Channel	Monitored point	Note
1, 2	Cavity A window	Two fibers per cavity window
3, 4	Cavity B window	Two fibers per cavity window
5, 6	Cavity C window	Two fibers per cavity window
7, 8	Cavity D window	Two fibers per cavity window
9	Klystron output window	
10	Circulator	
11, 12	(empty slots)	Two spare detector modules held on the shelf

12 detector modules are procured. Ten are installed and assigned as above; the remaining two are shelf spares, and their chassis slots stay empty.

12.5 Signal Path and Design

The chain is **optical fiber → ARC4 detector module → ARC4 backplane → ARC4 system modules → Interface Chassis**, shown in Figure 4.

- **Fast trip path**: a detector module that sees light asserts on the backplane, and the system modules combine all detector states into the arc permit presented to the Interface Chassis. The permit is active-high and the whole path is hardware — no processor, no encoding step — so the trip reaches the Interface Chassis AND-gate in the same microsecond class as the LLRF9's own FPGA interlocks.
- **Diagnostic identification path**: the identity of the detector that fired is held inside the ARC4 and presented to the Interface Chassis alongside the permit. The Interface Chassis includes it in the fault status word that the RF MPS PLC reads, so the first-fault record names the specific window rather than only "arc".
- **Test path**: the test module lets the complete chain — detector module, backplane, system module, Interface Chassis permit input — be exercised without a real arc. This is what makes a periodic proof test of the arc interlock possible, which the legacy Fast IC arc inputs never supported.

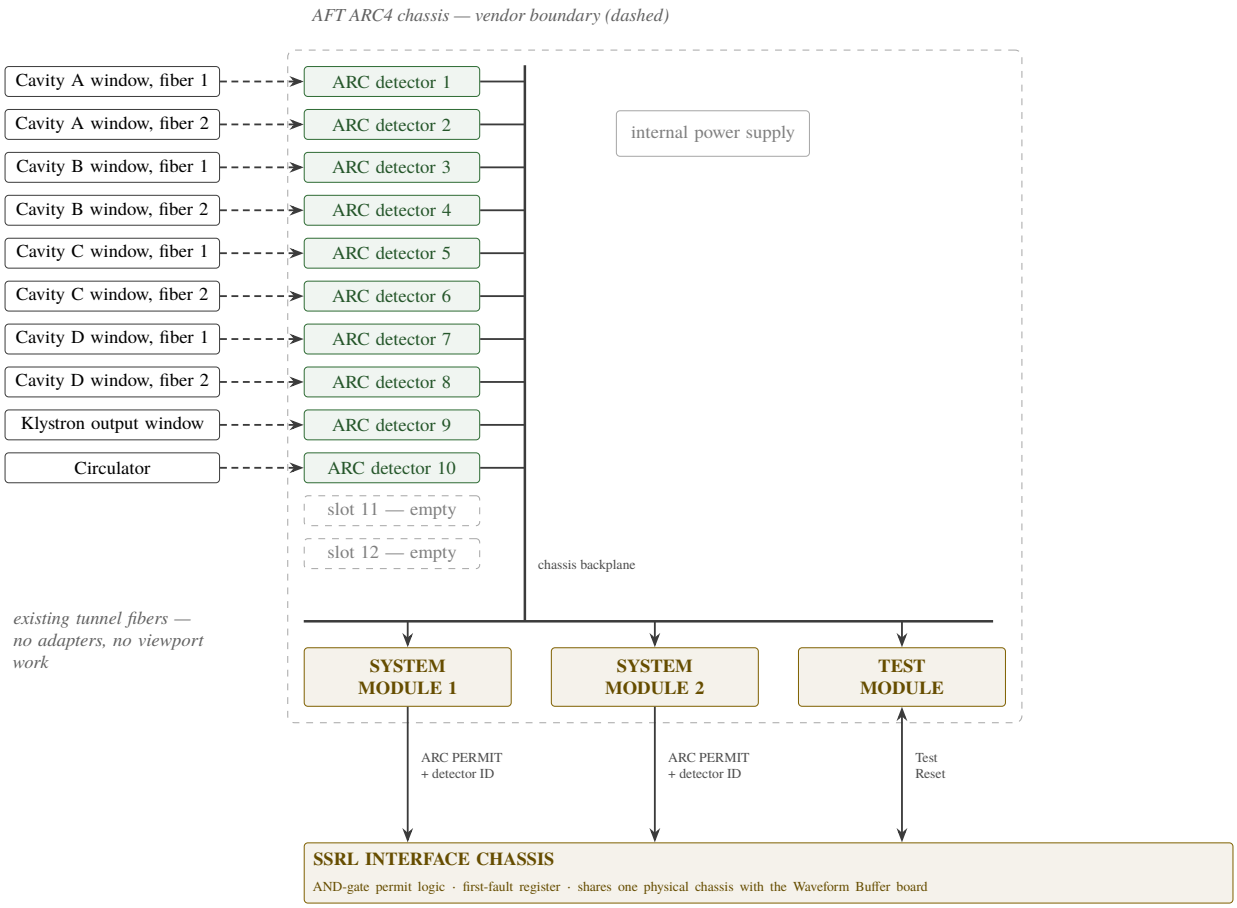


Figure 4: Arc detection signal path. The dashed outline is the vendor boundary: everything inside it is AFT ARC4 hardware, and no SSRL wire lands on an individual detector module. Fibers enter the detector modules from the front; the detector modules drive the chassis backplane; only the two system modules and the test module face the Interface Chassis. Ten of the chassis's twelve slots are populated; slots 11 and 12 are left empty and their two detector modules are held as shelf spares.

Open item — ARC4 external interface definition. The exact electrical form of the system-module and test-module interface is not yet fixed: how many discrete lines there are, their logic levels, whether the fired-detector identification arrives as parallel bits or as an encoded word, and how the latch is reset. These must be taken from the AFT ARC4 documentation and settled **before the Interface Chassis PCB is laid out**. [O5]

12.6 Permit Logic

- The arc permit from the ARC4 system modules is **active-high** (permit present = no arc); any arc event drives it low — consistent with the fail-safe convention throughout the Interface Chassis
- The Interface Chassis removes the RF permit immediately upon any arc event
- The fired-detector identification carried in the Interface Chassis fault status word indicates which channel fired, and is held until reset
- If multiple detectors fire simultaneously, all are reported
- Reset requires an explicit EPICS command via the RF MPS after the fault is acknowledged; the reset acts on both the ARC4 latch and the Interface Chassis first-fault register

12.7 Interfaces

- **Tunnel fibers → ARC4 detector modules**: one existing optical fiber per channel, landed on the detector module front panel. No adapters, no viewport changes.
- **ARC4 system modules (×2) → Interface Chassis**: arc permit (fail-safe active-high) plus fired-detector identification. Signal count and logic levels not yet defined — open item **O5**.
- **ARC4 test module ↔ Interface Chassis**: test initiate and test response, plus latch reset.
- **Interface Chassis → RF MPS PLC**: arc permit state and fired-detector identification included in the Interface Chassis fault status word.
- **RF MPS / EPICS**: per-channel trip status PVs, fired-detector identification, event count, latch reset command.
- **EPICS Coordinator**: reads arc fault status for operator displays and state machine fault handling.

Not an interface: nothing outside the ARC4 connects to an individual detector module. Adding or moving a channel is a change inside the vendor chassis only, and does not touch the Interface Chassis design.

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13 Subsystem 9: Klystron Cathode Heater

Detailed reference: `llrf/documentation/filamentHeater/sd3403110002.pdf`

13.1 Legacy System

The system was originally designed for the PEP-II B-Factory program at SLAC by P. Corredoura and the PEP-II LLRF group, and was subsequently adapted for use in the SPEAR3 storage ring RF system at SSRL. It has been in continuous service for over 25 years.

Key Design Characteristics:

Motor-driven variac providing continuously variable AC voltage

Solid-state relay (SSR) for on/off switching

Comprehensive monitoring: AC voltmeter, AC ammeter fed by a 10:1 current transformer (Texmate CT, T1), and elapsed-time meter

Remote control interface via Allen-Bradley (A/B) PLC digital/analog I/O

Front-panel indicators (DS1, DS2 green LEDs) and local metering

Power Supply Capabilities:

Parameter	Legacy
AC Input	120 VAC, 60 Hz
Maximum Power Rating	~1 kW (1000 W)
Nominal Operating Power	~500 W (actual sustained operation)
Nominal Operating Voltage	68 V (AC, chassis output)
Nominal Operating Current	7.3 A
RF MPS trip limits (PLC-5, live values)	62.4–76.2 V; 5.24–9.77 A
EPICS alarm limits LOLO / LOW / HIGH / HIHI (live values)	63.8 / 66.5 / 72.1 / 76.2 V; 6.72 / 7.01 / 7.59 / 7.88 A
Maximum Rating	140 V RMS, 1.0 kVA (chassis output)

13.2 Upgraded System

The **TDK Lambda programmable AC power supply** is the selected replacement for the motor-driven variac, and the unit is installed on a test stand, where it is to be accepted with a load. It requires an **external 60 Hz sync circuit**, which is being designed in-house by B. Morris, so that the heater output stays phase-locked to the site line rather than free-running against it. [O6]

Key advantages over the legacy variac:

- Analog voltage programming input — allows the RF MPS to directly ramp output voltage via a DAC signal, replacing the slow motor-driven variac
- Ethernet interface with an ASCII (SCPI-style) command set — enables EPICS monitoring and enable/disable control via an ASYN driver module
- Solid-state, no mechanical wear

Parameter	Legacy	Upgraded (TDK Lambda)
Control method	Motor-driven variac	Programmable AC supply, analog voltage ramp
Line synchronization	Inherent (variac on the line)	External 60 Hz sync circuit, in design (B. Morris)
Response time	Seconds–minutes	<100 ms
Voltage regulation	not specified	From the selected model's data sheet
Reliability	Mechanical wear	Solid-state; MTBF from the data sheet
EPICS integration	Limited	ASYN module over Ethernet (monitoring + enable/disable)

Control architecture:

- **MPS controls the supply** via an analog ramp on the programming input. The RF MPS PLC DAC output drives the TDK Lambda voltage programming pin, setting the output level. This keeps the safety-critical ramp control in hardware (MPS) with fast response.
- **External 60 Hz sync**: an in-house circuit (B. Morris) supplies the line-frequency reference to the supply. It is a new design item and is on the heater subsystem's critical path.
- **EPICS communicates via Ethernet** for monitoring and enable/disable. An ASYN EPICS module will be developed to provide readback PVs (output voltage, current, status) and to send enable/disable commands from the EPICS coordinator.
- **Existing V/I metering retained**: The current analog voltage and current meters (feeding MPS analog inputs) are kept in place to provide hardware-level monitoring of the output within rating limits, independent of the Ethernet path.

Key design considerations:

- MPS ramps the supply output voltage via analog programming input for controlled warm-up and cooldown; no mechanical components
- External 60 Hz sync circuit to be designed, built and tested (B. Morris) before the supply can be commissioned
- ASYN EPICS module required to interface to the TDK Lambda Ethernet port; module to be developed once the model and its command subset are fixed
- Existing analog V/I metering feeds MPS hardware inputs directly — hardware protection path is independent of Ethernet/EPICS
- Automated warm-up, standby, and cooldown sequences managed by the EPICS coordinator

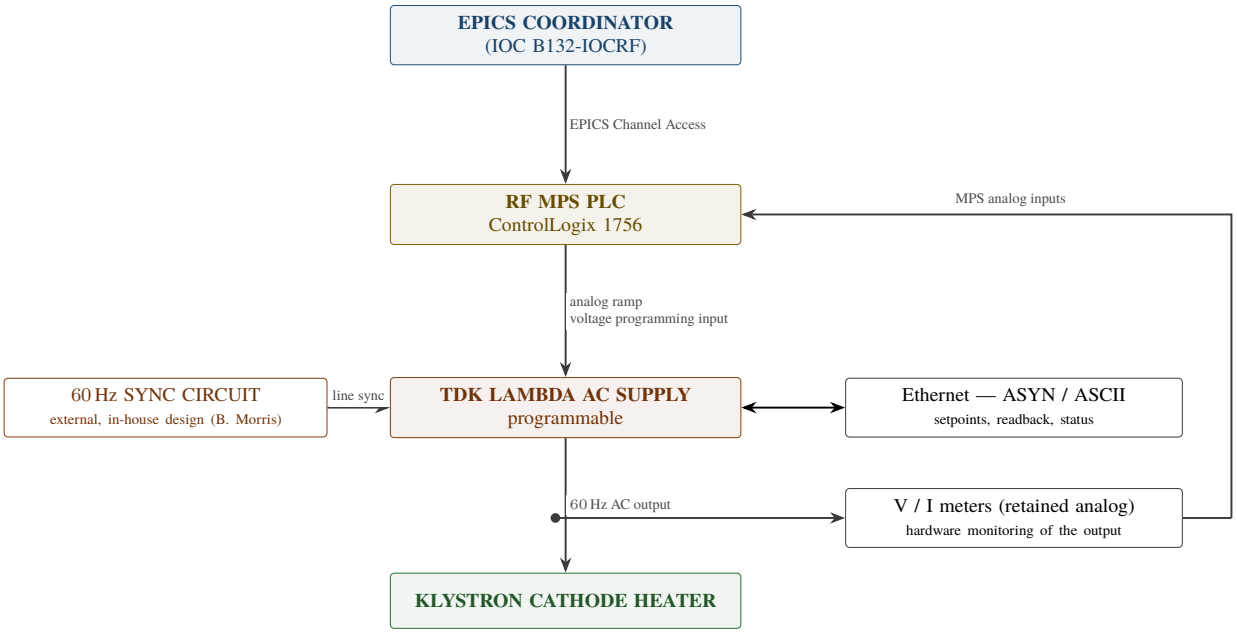


Figure 5: Klystron cathode heater controller architecture. Three paths are deliberately separate: the MPS ramps the TDK Lambda output through the *analog* programming input, so controlled warm-up and cooldown do not depend on the network; the Ethernet link carries setpoints and readback only; and the existing analog V/I meters are retained, so the delivered output is monitored in hardware independently of both. The external 60 Hz sync circuit is a new in-house design item. There are no mechanical components anywhere in the control path.

- Heater interlock: klystron does not receive an RF permit until the heater has reached operating power and the soak timer has elapsed; EPICS provides an expert override for short power dips

13.3 Interfaces

- **RF MPS → TDK Lambda (hardware control)**: MPS PLC DAC output drives the analog voltage programming input. Output is ramped up/down under MPS control for warm-up and cooldown sequences. MPS also provides the hardware enable/disable signal.
- **Sync circuit → TDK Lambda**: external 60 Hz line-frequency reference, in-house design (B. Morris).
- **EPICS → TDK Lambda (monitoring + soft control)**: ASYN EPICS module communicates over Ethernet using an ASCII command set. Provides readback PVs for output voltage, current, and status; sends enable/disable commands from the EPICS coordinator.
- **Existing V/I metering (retained)**: Legacy analog voltage and current meters feed MPS analog inputs directly, providing hardware-level protection independent of the Ethernet path. Output is monitored against rating limits.
- **HVPS coordination**: Heater must be at operating temperature before HVPS enable; HVPS must be off before heater cooldown.
- **RF MPS permit**: Heater “ready” status required for RF permit. Hardware heater current and voltage monitored by MPS analog inputs.

14 Subsystem 10: Control Software

Detailed reference: `spear-rf-code-legacy/rfApp/src/seq/`, `Docs_JS/LLRFOperation_jims.docx`

14.1 Legacy Software

The legacy control software consists of SNL (State Notation Language) programs running on VxWorks RTOS in the VXI crate processor:

Program	Lines	Function
`rf_states.st,v`	2,227	Master state machine (OFF/PARK/TUNE/ON_CW), turn-on, fault handling
`rf_dac_loop.st,v`	290	DAC control loop (gap voltage regulation)
`rf_hvps_loop.st,v`	343	HVPS control loop (drive power regulation)
`rf_tuner_loop.st,v`	555	Tuner control ($\times$4 cavities, phase-based, load angle)
`rf_calib.st,v`	3,345	Calibration (analog offset nulling, coefficient calibration, ~20 min)
`rf_msgs.st,v`	352	Message logging, HVPS faults, TAXI error detection
Total	**7,112**	Six programs, one `rfSeq` IOC library

Line counts are the head revision of each RCS archive in `spear-rf-code-legacy/rfApp/src/seq/`, counted including blank and comment lines.

14.2 Upgraded Software — EPICS/MATLAB Coordinator

> **Note**: Much of the new software functionality will already be in the LLRF9 internal EPICS IOC (written by Dmitry). Additional supervisory software will be in EPICS/Python/MATLAB as needed.

The upgrade replaces all SNL code with an EPICS/MATLAB coordinator application that operates as a supervisory control layer at ~1 Hz. It is NOT in the fast safety path.

14.3 Key Design Principles

- **Separation of concerns**: Each module has a single responsibility
- **Hardware abstraction**: All hardware interfaces go through the EPICS layer
- **Configuration-driven**: All operational parameters in configuration files
- **Fault tolerance**: Graceful degradation when subsystems are unavailable
- **Diagnostics**: Structured event logging, LLRF9 + waveform buffer readout, system performance metrics, and logging
- **Testability**: Mock interfaces for all hardware dependencies
- **Safety delegation**: Hardware safety handled by Interface Chassis; software handles sequencing and coordination only

15 Control Loop Architecture

This section maps the control loops in the system, identifying which hardware subsystem closes each loop, the loop bandwidth, and the upgrade path.

15.1 Fast RF Feedback (Direct Loop)

This is the fastest loop in the system and is entirely closed inside the LLRF9 FPGA. The EPICS coordinator does NOT participate.

15.2 HVPS Supervisory Loop (~1 Hz)

This loop trims the HVPS voltage setpoint to hold the requested operating point while respecting vacuum, gap-voltage and klystron-power limits. **The supervisory function is unchanged by the upgrade**; only the endpoints it reads and writes change.

The legacy implementation is `rf_hvps_loop.st,v` (343 lines), which runs a 0.5 s cycle with a 10 s hard deadline and operates in one of three states — `off`, `proc` and `on`:

- **PROC (cavity processing)**: each cycle the loop compares klystron forward power against its limit and checks the cavity vacuum and gap-voltage alarm severities. If any is bad it applies a fixed negative step to the voltage setpoint; if all are good it applies a fixed positive step. This is what walks the cavities up in voltage during conditioning.
- **ON (normal operation)**: the correction step is taken from the drive-power error when the direct loop is closed, or from the station gap-voltage error when it is not. The result is clamped to the configured voltage minimum and maximum.
- **Tolerance supervision**: if the voltage readback disagrees with the requested value by more than the allowed difference for 10 consecutive cycles, the loop holds the setpoint and raises a status code rather than continuing to drive. A separate 10-cycle counter guards the case where the cavities are already at their voltage limit.

The loop reports one of 16 status codes (for example `GOOD`, `CAVV_LIM`, `VOLT_TOL`, `VOLT_LIM`, `VACM_BAD`) together with a human-readable string, so the operator sees *why* the loop is holding rather than only that it is.

In the upgrade the same algorithm runs in EPICS, in the IOC B132-IOCRF, at ~1 Hz, with no Python layer and under the current HVPS PV names. The differences are all at the interfaces: the voltage setpoint and readback move over Ethernet to the CompactLogix HVPS PLC instead of through the Allen-Bradley serial link to the SLC-500, and the gap-voltage and drive-power inputs come from the LLRF9 units rather than from the RFP and IQA modules. Because the Waveform Buffer System now measures HVPS voltage and current directly, the loop also has a true power reading available where the legacy loop had only a forward-power proxy.

15.3 Tuner Control Loops (×4, ~1 Hz)

One loop per cavity nulls the cavity detuning by driving the tuner stepper motor until the probe-to-forward phase difference reaches its setpoint. **The control law is unchanged by the upgrade.**

The legacy implementation is `rf_tuner_loop.st,v` (555 lines), instantiated four times — once per cavity — with five states (`loop_init`, `loop_unknown`, `loop_off`, `loop_reset`, `loop_on`). Each cycle it:

1. reads the cavity phase error (probe phase minus forward phase);
2. converts it to a tuner position delta through the per-cavity gain;
3. clamps the resulting position to the motor drive limits and skips the move entirely if the change falls inside the deadband, so the tuner does not chatter;
4. commands the move and then waits for the motor's done-moving flag before trusting the new position.

Guards are on the motion rather than the arithmetic: five consecutive failed moves raise a motor-stall status, a missing or bad-severity phase measurement stops motion rather than moving on stale data, and forward power below the configured minimum disables the loop. There is a separate homing sequence with distinct home positions for the ON and PARK states, retried up to five times. The loop reports 14 status codes distinguishing the warning cases (at a drive limit, not at requested position, taking too long to move) from the non-functional cases (bad tuner readback, missing measurements, RF power too low).

In the upgrade the phase measurement comes from LLRF9 Unit 1, which delivers all four cavity probe and forward phases at 10 Hz with sub-degree resolution — an order of magnitude better than the legacy IQA chain. The loop is closed inside that unit, which drives the Galil DMC-4143 directly through its built-in Galil driver instead of the Allen-Bradley 1746-HSTP1. Its built-in tuner control PVs (`GAIN_P`, `GAIN_I`, `OFFSET`, `CLOSE`, `MINFWD` per cavity) set the PID gains, the load-angle offset, the loop state and the minimum forward power. The deadband, drive-limit and stall guards above are still required: the deadband and the minimum forward power are LLRF9 settings, and the drive limits are set in the Galil inside the mechanical travel of the tuners.

15.4 Load Angle Offset Loop (~1 Hz)

This outer loop balances gap voltage across the four cavities by biasing the individual tuner phase setpoints, as described in Section 10.4. It reads the four cavity probe amplitudes, computes the imbalance, and adjusts each cavity's phase offset to redistribute power.

Legacy status: the framework exists inside `rf_tuner_loop.st,v` and the offset is computed in the EPICS database, but it is *not used* on SPEAR3 — the phase offset fed to the tuner servo is fixed at zero, so the legacy tuners run in pure resonance-seeking mode. The mechanism is inherited from PEP-II, where it regulated cavity voltage in park mode.

In the upgrade the loop becomes a distinct supervisory module in the EPICS coordinator rather than logic buried in the tuner sequencer, writing to the per-cavity tuner phase offset. Whether it is enabled is an operational decision; if it is, it needs an integrator limit and a hold condition tied to tuner loop health, because an outer loop writing into a servo that is itself faulted is a known way to walk a tuner into a limit.

15.5 Station State Machine

The station sequencer owns turn-on, turn-off, mode changes and the response to a fault. **The sequencing responsibility is unchanged by the upgrade**; the state set is simplified because much of the legacy complexity existed to manage PEP-II loops that SPEAR3 does not use.

The legacy implementation is `rf_states.st,v` (2,227 lines) — by a wide margin the largest piece of legacy control software. It comprises 23 states across three concurrent state sets:

- **5 operating states**: `OFF`, `PARK`, `TUNE`, `ON_FM`, `ON_CW`.
- **12 transition substates** that carry the station between them, including the orderly shutdown, the station-reset retry, and the fault-file capture.
- **5 loop-protection substates** that ramp gap voltage up and down and ramp the direct and comb loop gains, plus a concurrent fault-file writer.

Not every transition is legal. `PARK` cannot go directly to either ON state — it must pass through `TUNE` — and returning from `ON_CW` to a lower state is routed through a controlled fallback rather than a direct jump. On a fault the sequencer captures the fault waveform files, performs an orderly shutdown, and then attempts an automatic restart up to a retry limit; a forced-fault latch blocks that automatic restart until an operator clears it.

In the upgrade the sequencer moves to the EPICS coordinator and the state set collapses, because `ON_FM` and the comb and gap-voltage-feedback ramps it existed to manage are PEP-II features that SPEAR3

does not use (Section 15.7). Ramping drive power and closing the field loop become LLRF9 operations; the HVPS turn-on and turn-off sequences become CompactLogix operations; tuner motion becomes an LLRF9 operation through its Galil driver (the legacy homing to fixed ON and PARK positions has no counterpart, so turn-on from a detuned tuner position is verified at commissioning). What the coordinator retains is exactly the part that was always supervisory: deciding which transitions are legal, ordering the subsystem operations, arming and clearing the fault response, and enforcing the retry limit and the forced-fault latch.

The one behaviour that must be preserved carefully is the interaction with the protection chain. The legacy sequencer could afford to be in the trip path because it ran on a real-time kernel; the upgraded coordinator is explicitly *not* in the fast safety path (Section 17.1), so the state machine must react to a trip that the Interface Chassis has already executed, rather than being the thing that executes it.

15.6 Calibration

Calibration nulls the offsets through the RF signal chain and measures the modulator response so that commanded I/Q corresponds to actual field. **The need is unchanged by the upgrade; the implementation is entirely replaced.**

The legacy implementation is `rf_calib.st,v`, 3,345 lines running a 28-state sequence that takes about 20 minutes. Each stage performs a binary search on a DAC channel, settling after each step, averaging a long I/Q record, and halving the step until the residual is within one LSB. It works through the cavity modulator terms, the direct loop, the comb loop, the summing and gain stages, the klystron stage and demodulator, and the final modulator null — on the order of sixty DAC channels. It aborts cleanly on operator interrupt, on a missing comb module, and when RF is not enabled, restoring the saved DAC values on the way out.

In the upgrade almost all of this disappears with the hardware it calibrated. The offsets being nulled were those of the analog RFP signal chain; the LLRF9 performs its own internal calibration of its ADC and DAC chains and provides it as a built-in function. What remains at the system level is the *station* calibration that no single unit can do for itself: establishing the phase and amplitude relationship between the two LLRF9 units, the cable-length and coupler-calibration constants that convert LLRF9 readings into cavity gap voltage and waveguide power, and the tuner phase reference per cavity. These are coordinator-level procedures operating on LLRF9 measurements rather than DAC-level nulling, and they should be expected to take minutes rather than the legacy twenty.

15.7 Eliminated Loops

The following legacy loops are eliminated in the upgrade because the LLRF9's digital feedback inherently provides their function (note: comb filter and GVF were used for PEP-II, not SPEAR3):

- **Ripple rejection loop** — LLRF9 digital feedback inherently rejects power-line ripple
- **Comb filter loop (CFM)** — used for PEP-II multi-bunch stabilization, not applicable to SPEAR3
- **Gap voltage feedback (GVF)** — used for PEP-II, not SPEAR3; cavity field stabilization now handled by LLRF9 vector sum feedback
- **4-way branching (DAC loop)** — Eliminated; LLRF9 controls all via single vector sum

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16 Inter-Subsystem Interface Matrix

This matrix summarizes all physical and logical interfaces between subsystems in the upgraded system.

From To	LLRF9 #1	LLRF9 #2	HVPS PLC	RF MPS PLC	Interface Chassis	Waveform Buffer	Arc Detect	Motor Ctrl	Heater Ctrl	EPICS Coord
LLRF9 #1	—	Interlock daisy (LEMO)	—	—	Status (5V digital)	—	—	Tuner moves (Ethernet)	—	EPICS (Ethernet)
LLRF9 #2	Interlock daisy (LEMO)	—	—	—	Status (5V digital)	—	—	—	—	EPICS (Ethernet)
HVPS PLC	—	—	—	—	STATUS (fiber), SCR EN (fiber), CROWBAR (fiber)	—	—	—	—	EPICS (Ethernet)
RF MPS PLC	—	—	—	—	Summary, Heartbeat, Reset (digital)	—	—	—	—	EPICS (Ethernet)
Interface Chassis	Enable (3.2V)	Enable (3.2V)	SCR EN, CROWBAR (fiber)	Fault status (digital)	—	—	—	—	—	—
Waveform Buffer	—	—	—	—	Comparator trips (digital)	—	—	—	—	EPICS (Ethernet)
Arc Detect	—	—	—	—	Permit + Latching	—	—	—	—	—
Motor Ctrl	Position, limits (Ethernet)	—	—	—	—	—	—	—	—	Via LLRF9 #1
Heater Ctrl	—	—	—	—	—	—	—	—	—	EPICS (Ethernet)
EPICS Coord	EPICS (Ethernet)	EPICS (Ethernet)	EPICS (Ethernet)	EPICS (Ethernet)	EPICS	EPICS (Ethernet)	—	Via LLRF9 #1	EPICS (Ethernet)	—

From To	LLRF9 #1	LLRF9 #2	HVPS PLC	RF MPS PLC	Interface Chassis	Waveform Buffer	Arc Detect	Motor Ctrl	Heater Ctrl	EPICS Coord
External (SPEAR MPS, Orbit)	—	—	—	—	24V permits	—	—	—	—	—

The Interface Chassis also drives the beam-abort output to the SPEAR MPS, which has no column in the matrix (Section 8.2).

16.1 Interface Signal Types Summary

Type	Used for	Between	Why this medium
Fiber optic	SCR ENABLE, CROWBAR, STATUS	Interface Chassis ↔ HVPS	Full galvanic isolation across a building boundary at high voltage; immune to the ground transients a crowbar event produces
Digital, opto-isolated	Permits, heartbeat, reset, fault status	Interface Chassis ↔ MPS, Waveform Buffer, Arc Detect	Deterministic, no protocol stack; response set by propagation not by scan rate
LEMO interlock	Interlock daisy-chain	LLRF9 #1 ↔ LLRF9 #2	Vendor-native interface; propagates a reflected-power trip between units without going through the Interface Chassis
Dry contact / solid-state relay	Arc detection outputs	AFT ARC4 system modules → Interface Chassis	Vendor-native chassis output; isolated contact closure needs no supply on the SSRL side
Analog	Heater voltage programming; HVPS voltage and current; V/I metering	MPS → TDK Lambda; HVPS → Waveform Buffer and MPS	Continuous quantity with a hardware path that stays valid if the network is down
EPICS Channel Access	Setpoints, readbacks, waveforms, status, commands	EPICS coordinator ↔ all intelligent devices	Supervisory only; every device on it holds its last safe state if the link drops

Two conventions apply across all of them. Every permit is **active-high**, so a broken cable, a lost supply or a dark fiber removes the permit rather than granting it. And every interlock signal that crosses a chassis

boundary is isolated — by optocoupler, by fiber, or by relay contact — so that no fault current or ground offset propagates between subsystems.

16.2 Open Interface Questions

Every unsettled interface detail is carried in the master **Open Items** table in the front matter (p. 7). The interface entries in that table are **O1** (LLRF9 $\leftrightarrow$ Interface Chassis logic levels), **O3** (RF Local Display Panel logic and beam-abort endpoint), **O4** (Waveform Buffer channel assignment), **O5** (AFT ARC4 $\rightarrow$ Interface Chassis), **O6** (heater supply) and **O7** (network addressing). Each must be closed before the corresponding subsystem can be integrated.

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17 Protection Chain and Interlock Architecture

17.1 Protection Philosophy

The upgraded system implements a layered protection architecture:

- **Layer 1 — LLRF9 FPGA interlocks** (few μs): RF overvoltage, baseband window comparators, internal housekeeping. These are the fastest protection and act within the LLRF9 itself to disable the DAC output.
- **Layer 2 — Interface Chassis hardware** (few μs from input change): Aggregates all permit signals and coordinates system-wide protection. All signals are optocoupler-isolated or fiber-optic. First-fault detection identifies the initiating event.
- **Layer 3 — RF MPS PLC** (~ms): Aggregates RF interlock status with external safety systems (SPEAR MPS, orbit interlock). Provides reset signal to Interface Chassis.
- **Layer 4 — EPICS Coordinator** (~1 s): Supervisory monitoring, logging, fault analysis, and recovery sequencing. NOT in the fast safety path.

17.2 Fault Propagation Example — RF Arc Event

17.3 Fail-Safe Design

All critical subsystems are designed to fail safe:

Subsystem	Failure Mode	Safe State
Interface Chassis power loss	All outputs de-energize	LLRF9 disabled, HVPS SCR disabled (K4/Ross controlled by separate PPS interface box)
LLRF9 power loss	Interlock status goes low	Interface Chassis removes permits
HVPS PLC failure	STATUS signal lost	Interface Chassis removes permits
RF MPS PLC failure	Heartbeat lost	Interface Chassis removes permits
EPICS coordinator crash	No effect on safety	Hardware protection continues; supervisory control paused
Ethernet failure	No effect on safety	Hardware protection continues; supervisory control paused

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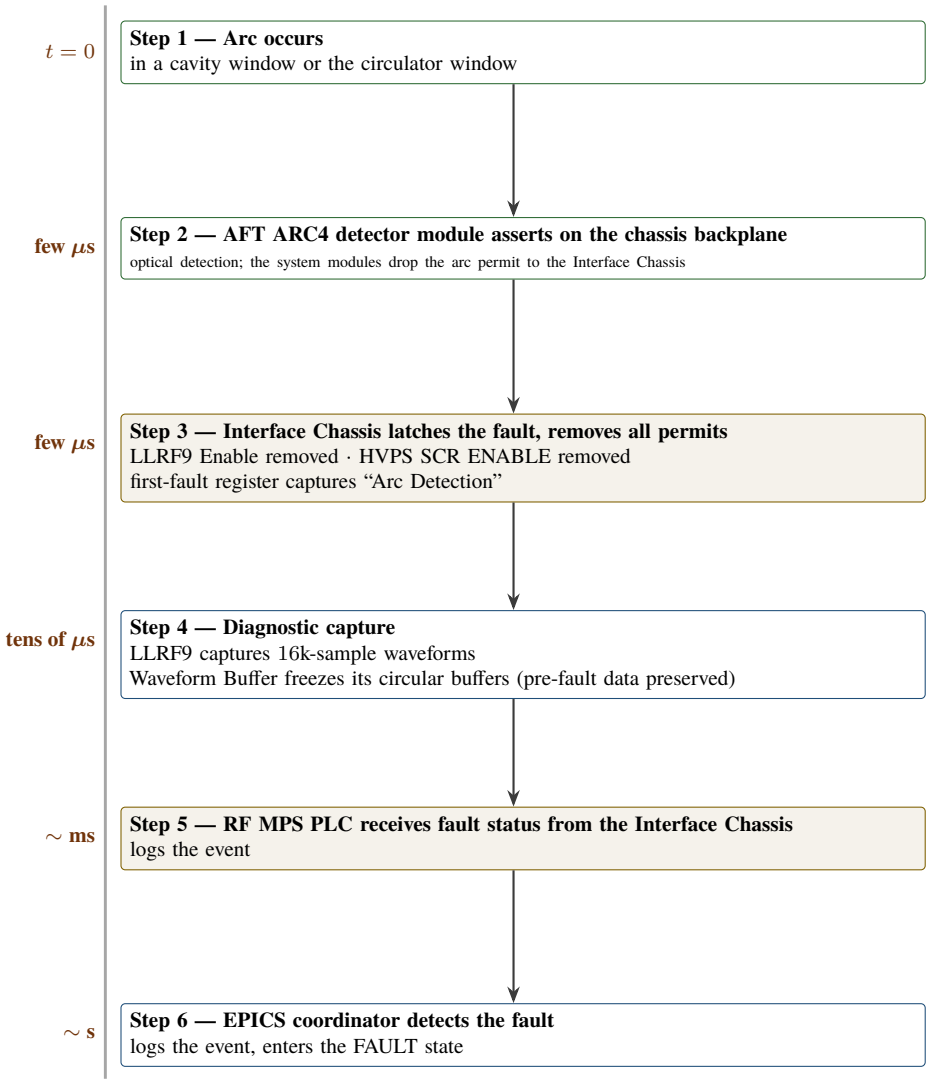


Figure 6: Fault propagation for an RF arc event. Steps 1–3 are pure hardware and complete within a few microseconds: protection is finished before any programmable device is involved. Each of the remaining steps is roughly three orders of magnitude slower than the one above it, and steps 4–6 are diagnostic and supervisory only — no protective action depends on them, so a PLC or IOC failure degrades observability but never safety.

18 Communication Architecture

18.1 EPICS Channel Access Network

All supervisory communication in the upgraded system uses EPICS Channel Access over Ethernet:

Device	IP Address	IOC Type	Key PV Prefix	Update Rate
LLRF9 Unit 1	TBD [O7]	Built-in Linux IOC	`LLRF1:`	10 Hz (scalars)
LLRF9 Unit 2	TBD	Built-in Linux IOC	`LLRF2:`	10 Hz (scalars)
HVPS CompactLogix	TBD	Served by the IOC B132-IOCRF	`SRF1:HVPS:` (current names)	~1 Hz
RF MPS ControlLogix	TBD	External EPICS gateway	`SRF1:MPS:`	~1 Hz
Motion Controller (Galil)	TBD	None; driven by LLRF9 Unit 1	Through `LLRF1:`	On demand
Waveform Buffer	TBD	Dedicated IOC	`SRF1:WFBUF:`	~1 Hz / event
EPICS/MATLAB Coordinator	TBD	IOC B132-IOCRF; caproto / PyEPICS / MATLAB clients	Various	~1 Hz

18.2 Non-EPICS Communication Paths

Path	Type	Purpose
Interface Chassis ↔ all hardware	Hardwired digital/fiber	Safety interlocks (no network dependency)
LLRF9 Unit 1 ↔ Unit 2	LEMO interlock daisy-chain	Fast interlock propagation
LLRF9 Unit 1 ↔ Galil DMC-4143	Ethernet, Galil command interface (LLRF9 built-in driver)	Tuner motion and readback
Interface Chassis → SPEAR MPS	Hardwired	Beam abort (station not ready for beam)
PPS ↔ PPS Interface Box	Hardwired (GOB1208PNE)	Personnel safety (no network dependency)
AFT ARC4 system modules → Interface Chassis	Dry contact relay and/or semiconductor switches	Arc interlock (no network dependency)

Critical design principle: All safety-critical communication paths are hardwired. No safety function depends on the Ethernet network or EPICS Channel Access. Network loss pauses supervisory control but does not compromise protection.

19 Implementation Phases and Risk Summary

19.1 Implementation Phases

Critical Constraint: The SPEAR3 LLRF upgrade has extremely limited flexibility for integration testing since bringing systems online directly impacts SPEAR operations. This severely constrains the available testing window and requires a fundamentally different approach than typical development projects.

Key Implications:
- Maximum standalone subsystem testing must occur before installation
- Integration testing must be incremental and carefully planned
- Some subsystems can be brought online for limited testing without full operational impact
- Full system integration testing window is minimal

19.1.1 Phase 1: Maximum Standalone Development

- **Software Framework**: Develop with simulated interfaces where live hardware unavailable
- **Tuner Testing**: Test Galil controller with Booster RF cavity
- **LLRF9 Installation**: Install both units, connect RF signals, and bring them online
- **RF MPS Online**: Bring RF MPS online for EPICS interface development
- **HVPS PLC Online**: Enables EPICS software development
- **Waveform Buffer Assembly**: Complete assembly and standalone testing — **done, April 2026**
- **Interface Chassis Design/Fab**: Complete design and fabrication (shares the Waveform Buffer chassis; PCB at fab since August 2026, bench test October 2026)
- **HVPS Controller**: Design, fabrication, and installation
- **Arc Detector**: Receive and bench-test the AFT ARC4 chassis and its 12 detector modules; define the system-module interface from vendor documentation. *No SSRL design or fabrication — this is procurement and integration.*
- **PPS Interface Box**: Design, fabrication, and commissioning
- **Test Stand 18**: Upgrade to work with the upgraded HVPS controller

19.1.2 Phase 2: Incremental Subsystem Integration

- **Tuner Test with SPEAR Cavity**
- **HVPS Integration**: Integrate HVPS PLC with EPICS coordinator (test at test stand T18)
- **RF MPS Integration**: Integrate RF MPS with LLRF9, waveform buffer, arc detector, HVPS PLC, interface chassis, and software coordinator with existing RF signals

19.1.3 Phase 3: Critical Path Integration

- **LLRF9 Integration — Basic RF Processing**: Verify vector sum, feedback loops, online calibration
- **EPICS Integration**: Configure IOC, verify all PV interfaces
- **End-to-End Interlock Testing**: Verify complete interlock chain (limited RF power)

19.1.4 Phase 4: Full Power Commissioning

- **Incremental Power Ramp**: Gradually increase power while monitoring all systems
- **Performance Validation**: Verify all success criteria are met
- **Operator Training**: Train operators on new system

- **Documentation**: Complete commissioning report

19.2 Transition: Integration Sequence

The upgrade replaces the legacy equipment inside the running station, one integration per scheduled down, in the order set by the Subsystem Integration and Migration Plan (Doc M, `Designs/tex/docM-integration-migration/`). Doc M gives for each integration the legacy interfaces it must keep, the hardware and software it needs, its tests, its entry criteria and its fallback; Phases 2 to 4 above are carried out within this sequence.

Step	Integration	Replaces
I0	Preparation	—
I1	AFT ARC4 arc detection and the two LLRF9 units, monitor-only	The disconnected legacy optical arc front end
I2	Interface Chassis and Waveform Buffer	Fast Interlock Chassis 340-308 and RF Local Display Panel 340-310
I3	Tuner control by LLRF9 Unit 1 and the Galil, with the RFP still driving	The 1746-HSTP1 tuner control
I4	RF MPS (ControlLogix)	The PLC-5
I5	Klystron heater supply under RF MPS control	The variac stage
I6	LLRF9 as drive source, in parallel with I4 and I5	The RFP as drive source
I7	New HVPS controller (Hoffman box) and PPS Interface Box	The legacy Hoffman box and its PPS wiring, one HVPS unit at a time
I8	Legacy retirement	The VXI crate, the Remote I/O link and the tuner chassis; the IOC B132-IOCRF moves to new hardware

Until I6 the legacy SNL programs stay in charge; the coordinator comes as a minimal release on the VXI CPU for I6 and as the full release on new hardware for I8. The legacy PV names are kept, and each legacy unit stays in place as the fallback until the next integration has run without it.

19.3 Status

Item	Status	Notes
LLRF9 (4 units)	Hardware complete	Two units installed in B132; RF cables to be connected in the coming weeks; Unit 1 will drive the Galil. Two spares on the shelf.
Galil DMC-4143 tuner controller	Built and tested	Tested on the SPEAR3 tuner motors under an EPICS control by M. Dunning; LLRF9 Unit 1 will drive it directly

Item	Status	Notes
RF MPS PLC modules	Hardware complete	EPICS software under development
HVPS PLC modules	Hardware complete (HVPS1, HVPS2, B44 Test Stand)	EPICS software under development
Enerpro FCOG1200 boards	On order	5 boards ordered; awaiting delivery
Arc detection (AFT ARC4)	Done	Chassis fully assembled with all modules, September 2026
Waveform Buffer System	**Complete**	April 2026; shares one chassis with the Interface Chassis
Interface Chassis	In progress	Prototype built. Front and back panels to be modified and the chassis raised from 2U to **3U** to arrange components better; bench test Oct–Nov 2026.
Heater Controller (TDK Lambda programmable AC supply)	Supply on a test stand	Acceptance with a load on the test stand; software specification outstanding; external 60 Hz sync circuit in design (B. Morris)
PPS Interface Box	Initial version designed	Documentation mostly done; needs ADSO approval and the final design
Software	In progress	Legacy code review complete; functional requirements being drafted; beta expected Spring 2027

19.4 Key Technical Risks

Risk	Severity	Status	Mitigation
Tuner motor controller reliability	High	**Closed**	Galil tested on SPEAR in August 2026 under an EPICS control by M. Dunning, driving all four cavity tuner motors simultaneously; in service LLRF9 Unit 1 drives it directly
HVPS PLC code migration (SLC-500 → CompactLogix)	High	In progress	Reverse-engineer legacy code; build Test Stand 18 (B44)
PPS Interface approval delays	High	In progress	Engage protection managers early; provide complete documentation. Initial version designed and documentation mostly done; ADSO approval and the final design outstanding.

Risk	Severity	Status	Mitigation
Interface Chassis logic design (LLRF9/HVPS feedback loop)	High	In progress	Design and simulate before fabrication; careful sequencing logic. Prototype built; panels being reworked and the chassis raised to 3U; bench test Oct–Nov 2026.
Waveform Buffer System (new custom hardware)	Medium	**Closed**	Staged development completed; board complete April 2026
EPICS/MATLAB Coordinator software	Critical	In progress	Largest software scope: state machine, fault handling, auto-recovery sequences, operator interface, and integration with all subsystem IOCs. Legacy code review complete and functional requirements being drafted; beta expected Spring 2027. Must be developed in parallel with hardware integration.
RF MPS PLC software integration	High	In progress	MPS hardware is assembled and tested standalone; EPICS IOC development is under way, but integration with Interface Chassis fault status, first-fault logic and reset sequencing has not begun.
AFT ARC4 external interface definition	Medium	In progress	No longer an SSRL firmware task — the OR and latch functions are inside the vendor chassis. The residual risk is that the system-module and test-module interface must be pinned down from AFT documentation before the Interface Chassis PCB is laid out.
Heater supply 60 Hz sync circuit	Medium	In progress	TDK Lambda supply is installed on a test stand; the sync circuit design, build and test (B. Morris) is a new in-house item on the heater critical path, and the software specification is still outstanding.
Communication latency (Ethernet/EPICS for ~1 Hz control)	Low	Validated	Proven in LLRF9 prototype commissioning

19.5 Success Criteria

Metric	Legacy Performance	Target
Amplitude stability	<0.1%	Same or better
Phase stability	<0.1 deg	Same or better
Tuner resolution	~0.002–0.003 mm/microstep	Improved (up to 256 microsteps/step)
Control loop response	~1 second	Same or better
Uptime	>99% (nominal, not formally measured)	Same or better

Metric	Legacy Performance	Target
Fault diagnostics	Limited fault file capture	16k-sample waveform + circular buffer + first-fault

20 Appendix: Source Document Index

Every path below has been checked to exist. Note that the legacy SNL source is RCS-archived under `spear-rf-code-legacy/rfApp/src/seq/`, so each file carries a `.st,v` extension — there is no `llrf/legacyLLRF/` directory.

20.1 Governing project document

Path	Content
`Designs/docx/SPEAR3_LLRF_PDR_R2.docx`	**SPEAR3 LLRF Upgrade Preliminary Design Review, R2 (28 April 2026)** — the authoritative upgrade specification. Sections 1–19 of this report track it directly
`Designs/docx/SPEAR3_LLRF_Upgrade_Project_Schedule.xlsx`	Project schedule

20.2 LLRF Source Material

Path	Content
`../spear-rf-code-legacy/rfApp/src/seq/rf_states.st,v`	Legacy master state machine (23 states)
`../spear-rf-code-legacy/rfApp/src/seq/rf_dac_loop.st,v`	Legacy DAC control loop
`../spear-rf-code-legacy/rfApp/src/seq/rf_hvps_loop.st,v`	Legacy HVPS control loop
`../spear-rf-code-legacy/rfApp/src/seq/rf_tuner_loop.st,v`	Legacy tuner control loop
`../spear-rf-code-legacy/rfApp/src/seq/rf_calib.st,v`	Legacy calibration system
`../spear-rf-code-legacy/rfApp/src/seq/rf_msgs.st,v`	Legacy message logging
`../spear-rf-code-legacy/codeReviewTechnicalNotes/`	Nine-part code review of the legacy application
`llrf/architecture/llrfInterfaceChassis.docx`	Interface Chassis specification
`llrf/architecture/WaveformBuffersforLLRFUpgrade.docx`	Waveform Buffer System design
`llrf/architecture/arcDetectorHardwareOptions.docx`	Arc detection hardware selection
`llrf/architecture/rfPowerDetector.docx`	RF power detector selection
`llrf/architecture/analogDesignComponents.docx`	Analog component selection
`llrf/documentation/LLRFOperation_jims.docx`	SPEAR3 RF Station Operation Guide (J. Sebek)
`llrf/documentation/LLRFUpgradeTaskListRev3.docx`	Full project scope, procurement, costs

Path	Content
`llrf/documentation/fiberOpticCableSignalControlRev3.docx`	Fibre-optic signal and control routing
`../llrf/tuners/galil/`	Galil DMC-4143 firmware and **test-move logs** (the controller is *not* installed — see §10)
`../llrf/driveAmp/`	Drive amplifier datasheet (KAW2051M12)
`../llrf/arcDetector/`	Arc detector product sheets and mechanical references
`llrf/tests/llrf9Tests.tex`	LLRF9 commissioning measurements (J. Sebek, 2021)

20.3 HVPS Source Material

Path	Content
`hvps/documentation/wiringDiagrams/ei7307900000.pdf`	**EI-730-790-00-C0 / NWL #39308** — the transformer manufacturer's own schematic. Authoritative for the **2750 kVA / 127 A** nameplate and all oil, pressure, vacuum and flow protection setpoints
`hvps/documentation/schematics/sd7307900101.pdf`	SD-730-790-01-C1 — HVPS system schematic
`hvps/documentation/wiringDiagrams/wd7307940400.pdf`	WD-730-794-04-C0 — contains the **HVPS output voltage dividers**
`hvps/documentation/wiringDiagrams/wd7307900206.pdf`	WD-730-790-02-C6 — master trigger enclosure wiring diagram
`hvps/documentation/procedures/spear3HvpsHazards.tex`	J. Sebek — stored energy, discharge time constants, klystron perveance
`hvps/architecture/designNotes/EnerproVoltageandCurrentRegulatorBoardNotes.docx`	J. Sebek — authoritative regulator board circuit analysis
`hvps/architecture/designNotes/RFSystemMPSRequirements.docx`	RF system MPS requirements
`hvps/architecture/originalDocuments/slac-pub-7591.pdf`	Cassel & Nguyen, PAC 1997 — crowbar timing, regulation, ripple
`hvps/architecture/originalDocuments/ps3413600102.pdf`	PS-341-360-01 — HVPS power section specification
`hvps/controls/enerpro/enerproBoardHvps.docx`	J. Sebek — installed Enerpro board revisions, serials, phase references
`hvps/documentation/plc/CasselPLCCode.pdf`	Full SLC-500 ladder logic listing, program SSRLV6-4-05-10
`hvps/documentation/plc/plcNotesR1.docx`	J. Sebek — PLC timing and the N7:10 setpoint filter

Path	Content
`../hvps/documentation/procedures/`	Maintenance and safety procedures, EWPs, lockout permits
`../hvps/maintenance/`	Stack installation checklists, phase tank maintenance

20.4 Switchgear and PPS Source Material

Path	Content
`hvps/documentation/switchgear/gp3085000103.pdf`	**GP-308-500-01-R3** (LBL, 1977) — original design drawing; authoritative **relay legend**; blocking relay blocks trip above 2000 A
`hvps/documentation/switchgear/gp4397040201.pdf`	GP-439-704-02-C1 — as-built schematic and sequence of operation
`hvps/documentation/switchgear/id3088010601.pdf`	ID-308-801-06-C1 — as-built connection wiring
`hvps/documentation/switchgear/rossEngr713203.pdf`	Ross 713203 E-1 — HCA-1-A driver and HQ3 contactor
`pps/HoffmanBoxPPSWiring.docx`	J. Sebek — terminal-by-terminal PPS wiring trace
`pps/MSGfromJimSebektoFayaaboutPPS.md`	2022 PPS concerns and upgrade drivers
`pps/diagrams/README.md`	PPS diagram note set and the PPS readback trace